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Expert System for Road Surface Hazard Detection

A Deep Learning-Based Detection and Maintenance Prioritization Pipeline

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Abstract

Road infrastructure monitoring is essential for ensuring transportation safety, economic stability, and sustainable urban mobility. The increasing growth of traffic volume and aging pavement structures intensifies the need for scalable and reliable inspection systems. Traditional manual inspection methods are costly, time-consuming, and prone to subjective evaluation inconsistencies. Consequently, automated computer vision-based approaches have emerged as promising alternatives for large-scale road damage assessment.

Despite significant progress in deep learning-based road damage detection, most existing approaches focus primarily on maximizing detection accuracy on benchmark datasets. These systems often treat damage detection as an isolated perception task without addressing how detected defects should be prioritized for maintenance planning. As a result, raw detection outputs provide limited support for real-world infrastructure management decisions. This gap highlights the need for integrated systems that combine robust detection with interpretable decision-support mechanisms.

This thesis proposes a hybrid pipeline that integrates a YOLO-based deep learning object detection model with a rule-based expert system for hazard prioritization. The detection component identifies and localizes road surface defects from monocular RGB images under unconstrained environmental conditions. The expert system processes structured detection outputs and assigns maintenance priority levels based on defect type, confidence, and contextual criteria. The system is implemented using controlled experimental configurations to ensure reproducibility and transparent performance evaluation.

Experimental results demonstrate that lightweight YOLO-based configurations achieve competitive mean Average Precision while maintaining efficient inference performance. Class-wise analysis reveals robust detection of dominant crack categories while highlighting challenges in small-object and minority-class sensitivity. The integration of rule-based prioritization enables structured categorization of detected hazards into actionable maintenance levels. This separation between perception and reasoning improves interpretability without sacrificing detection quality. Controlled benchmarking ensures reliable comparison across configurations and strengthens reproducibility. The proposed modular design supports scalability and future architectural updates without restructuring the entire pipeline.

Overall, this research demonstrates that combining deep learning-based detection with explicit rule-based prioritization enhances both technical performance and operational relevance in road infrastructure monitoring. The proposed framework bridges the gap between automated visual inspection and actionable maintenance decision support in intelligent transportation systems.

Zusammenfassung

Die Überwachung von Straßeninfrastruktur ist von zentraler Bedeutung für Verkehrssicherheit, wirtschaftliche Stabilität und nachhaltige urbane Mobilität. Das zunehmende Verkehrsaufkommen sowie alternde Fahrbahnstrukturen erhöhen den Bedarf an skalierbaren und zuverlässigen Inspektionssystemen. Traditionelle manuelle Inspektionsverfahren sind kostenintensiv, zeitaufwendig und anfällig für subjektive Bewertungsunterschiede. Automatisierte, bildbasierte Verfahren bieten daher eine vielversprechende Alternative für großflächige Straßenzustandsanalysen.

Obwohl Deep-Learning-basierte Ansätze in der Straßenschadenerkennung erhebliche Fortschritte erzielt haben, konzentrieren sich viele bestehende Arbeiten primär auf die Maximierung der Erkennungsgenauigkeit auf Benchmark-Datensätzen. Die Frage, wie erkannte Schäden für Instandhaltungsentscheidungen priorisiert werden sollen, bleibt häufig unberücksichtigt. Rohdetektionen liefern ohne strukturierte Entscheidungslogik nur begrenzten Mehrwert für reale Infrastrukturverwaltungen. Daraus ergibt sich die Notwendigkeit integrierter Systeme, die robuste Erkennung mit nachvollziehbarer Entscheidungsunterstützung kombinieren.

Diese Arbeit entwickelt eine hybride Pipeline, die ein YOLO-basiertes Objekterkennungsmodell mit einem regelbasierten Expertensystem zur Priorisierung von Straßenschäden kombiniert. Die Detektionskomponente identifiziert und lokalisiert Fahrbahnschäden anhand von monokularen RGB-Bildern unter realen Umweltbedingungen. Das Expertensystem verarbeitet die strukturierten Detektionsergebnisse und weist auf Grundlage von Schadensart, Konfidenzwert und kontextuellen Kriterien Wartungsprioritäten zu. Die Implementierung erfolgt unter kontrollierten experimentellen Bedingungen zur Sicherstellung von Reproduzierbarkeit und transparenter Leistungsbewertung.

Die experimentellen Ergebnisse zeigen, dass leichte YOLO-Konfigurationen eine wettbewerbsfähige mittlere Average Precision bei gleichzeitig effizienter Inferenzgeschwindigkeit erreichen. Eine klassenweise Analyse verdeutlicht eine robuste Erkennung häufiger Risskategorien sowie bestehende Herausforderungen bei kleinen oder seltenen Defekten. Die Integration der regelbasierten Priorisierung ermöglicht eine strukturierte Einordnung der erkannten Schäden in umsetzbare Wartungskategorien. Die klare Trennung zwischen Wahrnehmung und Entscheidungslogik verbessert die Interpretierbarkeit ohne Einbußen bei der Detektionsqualität. Kontrollierte Benchmark-Experimente gewährleisten eine verlässliche Vergleichbarkeit der Modellkonfigurationen. Das modulare Systemdesign unterstützt Skalierbarkeit und zukünftige Erweiterungen.

Insgesamt zeigt diese Arbeit, dass die Kombination aus Deep-Learning-basierter Detektion und expliziter regelbasierter Priorisierung sowohl die technische Leistungsfähigkeit als auch die praktische Anwendbarkeit in der Straßeninfrastrukturüberwachung verbessert. Der entwickelte Ansatz schließt die Lücke zwischen automatischer visueller Inspektion und handlungsorientierter Instandhaltungsplanung.

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1 Introduction

The quality of road infrastructure greatly influences not just transportation safety, but also economic development and social interaction. As urban areas continue to grow and more vehicles take to the streets, it becomes increasingly important to find inspection and maintenance strategies that are both efficient and scalable for large networks. Traditionally, road inspections have relied on manual processes, which are often time-consuming, costly, and subject to individual interpretation. However, recent advancements in computer vision and artificial intelligence have given rise to automated systems that can examine road surfaces and detect defects with notable precision. These innovative solutions have proven to be highly effective in increasing the frequency and scope of inspections, while also bringing down operational costs [1]. Because of these advantages, the field of automated road damage detection has become increasingly relevant within the context of smart infrastructure and intelligent transport systems. By adopting these new solutions, it becomes possible to assess pavement conditions more objectively over different geographic locations. In addition, the shift to digital management of infrastructure brings the need for strong algorithms that are capable of performing well in various, less controlled environments. With the availability of extensive datasets, it is now possible for researchers to develop models that can adapt to unfamiliar road surfaces and changes in lighting.

Practical road maintenance involves more than just identifying issues on the road surface; it also demands effective prioritization of these issues based on their severity and potential impact. Contemporary smart maintenance systems now frequently utilize data-driven approaches to support both planning and the allocation of resources. Recent research emphasizes that combining perception technologies with advanced reasoning capabilities is essential for generating actionable insights, rather than merely noting the presence of defects [2]. Such comprehensive strategies contribute to infrastructure sustainability by enabling preventative maintenance actions and lowering future repair expenses. Building on these developments, this thesis presents a computer vision pipeline designed for detecting road hazards, alongside a prioritization framework that uses rule-based logic. The system is crafted to achieve an optimal balance between detection accuracy, interpretability, and real-world functionality during road inspections. By translating raw sensor data into practical decision-making tools, the framework gives municipal agencies a holistic resource for road management. Its modular design ensures that individual elements can be readily updated as new advances in deep learning become available. The overarching objective of this work is to demonstrate that automated inspection systems can achieve, and sometimes surpass, the consistency and speed of human evaluators.

1.1 Motivation

Ensuring the quality of road infrastructure is essential for supporting urban transportation, promoting road safety, and sustaining economic growth. As cities develop and traffic intensifies, surface issues such as cracks, potholes, and other forms of damage become increasingly frequent and severe. These problems not only compromise ride comfort and increase vehicle maintenance costs, but also raise the likelihood of accidents and place a significant financial burden on government bodies responsible for road upkeep. Historically,

inspections have depended on manual methods, which are both time-consuming and labor-intensive, making it difficult to maintain consistent assessments over large regions. Human judgment during inspection processes often introduces inconsistencies, leading to unreliable maintenance records and inefficient resource allocation. This situation highlights the need for technological solutions capable of providing reliable and repeatable road quality assessments. Automated monitoring enables more frequent evaluations, which are important for identifying and addressing pavement issues in time. Moreover, reducing dependence on manual inspections improves worker safety by limiting exposure to risks in busy traffic environments. Overall, the adoption of digital technologies is important for modern cities aiming to maintain efficient and resilient transportation infrastructure.

Recent advances in computer vision and deep learning have enabled automated systems that evaluate road conditions using visual data. These technologies allow wider and more frequent inspection coverage while reducing the costs of traditional methods. Despite these advantages, many existing solutions focus mainly on detection accuracy and do not address the decision-making required in real maintenance processes. In practice, road authorities need not only to detect defects, but also to prioritize them based on severity, urgency, and potential impact on safety and infrastructure lifespan [3]. In the absence of a clear prioritization framework, maintenance teams can be exposed to a large number of detection results that are not well organized, which makes planning and response difficult. The complexity of road environments makes this even more challenging, as changes in lighting and surface texture may negatively affect the performance of models that rely only on visual data. Domain knowledge should therefore be included in the evaluation, as it helps transform detection outputs into more practical and usable insights. The benefits of hybrid methods in addressing the limitations of black-box models are the main motivation behind this research. The proposed approach aims to meet the real-world needs of infrastructure asset management by focusing on both accuracy and interpretability.

Under these limitations, this thesis introduces a methodology that combines deep learning-based object detection with rule-based analysis derived from expert knowledge, in order to support interpretable and actionable prioritization in road maintenance. This approach aims to connect automatic defect detection with practical infrastructure management by combining data-driven perception models with decision-making based on domain-specific rules. It is increasingly accepted that this type of hybrid solution is suitable for public infrastructure applications, especially in contexts where safety and limited resources are important factors [4]. The inclusion of rule-based logic helps ensure that decisions can be understood and reviewed by human professionals, which is important for building trust with public agencies responsible for maintenance planning. In addition, the system has a flexible architecture that allows local policy requirements to be incorporated into the prioritization process. With the continued development of artificial intelligence, combining neural networks with symbolic reasoning is expected to play an important role in intelligent infrastructure systems. This thesis contributes by presenting a structured approach for road hazard assessment and a prototype that can support future smart city applications, where transparency and effectiveness are important.

1.2 Objective

This thesis looks at the design, implementation, and evaluation of an automated system for road damage detection using deep learning and interpretable decision logic. By combining object detection models with rule-based reasoning, the approach goes beyond simple detection and supports maintenance decisions.

In this work, a neural network architecture is selected to handle difficult real-world condi-

tions in road imagery. A rule-based expert system is also developed to reflect maintenance practices and safety aspects, allowing detected defects to be prioritized in a consistent way. The evaluation looks at both detection performance and the consistency and interpretability of the prioritization logic.

A key goal is that the system can be reproduced, scaled, and used in practical scenarios. It is tested under different road conditions and also in limited-resource settings to assess its feasibility. The trade-offs between model complexity, detection performance, and decision interpretability are analyzed, which is important for applied computer vision in smart infrastructure projects [5, 6]. All configurations and preprocessing steps are recorded to support reproducibility.

The analysis also looks at performance on data from different regions and the efficiency of the model for deployment on mobile or edge devices. Looking at these aspects gives a clearer view of how the system can be applied in real scenarios and how it can support more reliable and interpretable infrastructure inspection.

1.3 Problem Statement

Maintaining road infrastructure remains a challenge, especially when it comes to scalability, objectivity of inspections, and timely response. Traditional manual inspections are still widely used, but they are expensive, infrequent, and subject to human variability, which leads to inconsistent evaluations across inspectors. Although automated visual inspection systems have been proposed, many approaches focus mainly on detecting defects without addressing how these detections can be turned into prioritized maintenance actions. Because of this, such systems provide limited support for practical decision-making in real-world maintenance processes. In addition, many deep learning models work as black-box systems, which can reduce trust among stakeholders. Therefore, there is a need for systems that not only detect road defects but also provide interpretable and structured prioritization of maintenance actions.

Detecting road damage in real-world conditions introduces additional challenges, such as variations in lighting, weather, pavement types, and strong class imbalance. Models trained only for detection are often not transparent and do not clearly explain how predictions are made, especially under operational constraints. In addition, rare but critical defects are often underrepresented in available datasets, which makes them harder to detect reliably. These limitations show the need for an approach that combines accurate visual perception with domain knowledge to support clear prioritization.

To address these issues, this thesis proposes a hybrid approach that combines deep learning-based detection with rule-based reasoning. The goal is to link detection with decision-making by combining data-driven predictions with interpretable decision logic. The approach also aims to reduce false positives caused by visual ambiguities such as shadows, lane markings, and surface textures, while improving robustness when dealing with rare and critical defects. By combining visual detection with structured reasoning, the system aims to better align detection outputs with the needs of infrastructure maintenance [7, 8]. This work looks at how such an approach can improve reliability and practical use in automated road hazard assessment systems.

1.4 Thesis Scope

This thesis looks at the design, implementation, and evaluation of an automated system for road hazard detection and maintenance prioritization. The system works only with monocular RGB images and does not use additional sensors such as LiDAR or radar. It

focuses on common road defects like cracks, potholes, and surface degradation, using publicly available datasets. Different environmental conditions are also considered, especially changes in lighting and weather, since these are important in large-scale road monitoring.

Working with RGB images keeps the system simpler and more affordable, as it can be deployed using standard camera equipment. The RDD2022 dataset is used as a benchmark to allow consistent evaluation and easier comparison of results. Other sensing approaches can provide more information, but this work shows that a vision-based solution can still perform well when the modeling and evaluation are properly handled. The goal is to develop a system that is both technically and economically feasible and that can be applied in real infrastructure monitoring scenarios.

In addition to defect detection, this thesis incorporates a rule-based expert system that transforms model outputs into structured maintenance priorities. The focus is placed on system-level design, reproducibility, and the integration of decision-support logic, rather than on extensive modifications of neural network architectures. As a result, aspects such as multi-sensor fusion, real-time deployment on embedded hardware, and large-scale fleet integration are considered outside the scope of this work. The proposed approach aims to demonstrate how automated road inspection systems can be enhanced through the combination of data-driven perception and interpretable decision-making [9, 10].

The scope of this study is constrained by the limitations of available benchmark datasets, including restricted annotation quality and class distribution. No additional data collection or manual annotation was performed. Instead, the focus is on maximizing the effectiveness of existing data through controlled experimentation and systematic system design. These constraints define the boundaries of the study and provide a realistic assessment of system performance under practical conditions.

1.5 Purpose and Goal

In addition to the primary objective that we have discussed above, this part underscores the practical objectives, as well as what the proposed system is expected to contribute to. The project will reduce the time-consuming, expensive, and subjective human inspections of roads, as it will be done with deep-learning based object detection models. We are attempting to transform the crude visual information into properly organized data that can support infrastructure maintenance choices a step that aligns with the current developments in the intelligent transportation systems and digital infrastructure stewardship. This study is essential since metropolises across the globe are operating under constrained budgets in the process of maintaining old-fashioned infrastructure in decent condition. Providing a more effective inspection system favors the overall purpose of creating more sustainable urban systems that are safer. The paper also emphasizes the significance of ensuring that AI systems can be interpreted when applied to the service of the population. In the end, it is planned to map a path to the introduction of AI into the conventional engineering processes, a combination of which is fundamental in dragging the digital transformation of the infrastructure sector into practice.

The key point of the thesis is to design, implement, and test a hybrid system that combines both data-driven perception and rule-based decision logic. In practice, the system evaluates multiple object detection configurations with a variety of object detector setups, selecting an empirically appropriate model and integrating its output into a rule-based expert system, which gives the results of the maintenance priorities in accordance with the properties of the defects. The work also ensures that the work is reproducible and transparent, and of practical use, so I am adhering to standardized evaluation procedures and concentrating on system-level behaviour rather than focusing solely on isolated model

performance. Ultimately, the results demonstrate that the hybrid AI systems can solve the problem of high-performance visual detection and actionable decision support in real-world road maintenance as demonstrated by the sources such as [11, 12]. The success will provide research community with a proven model on how to conduct future projects and the road authorities with a tool which they can customize to enhance their maintenance planning. This piece of work, however, contributes to the larger discipline of intelligent systems, and the final result will be the comprehensive analysis of the potential and restrictions of the vision-based road hazard detection.

1.6 Outline

This thesis is divided into seven major chapters that address various aspects of the study and solution proposed. The design will guide the reader towards the theoretical basis all the way up to the practical implementation and the testing of the system.

Briefly, the work seals the research gap by providing a reproducible modular hybrid framework which distinctly divides the perception and the decision logic. Combining regulated YOLO-based detection benchmarking with a rule-based expert prioritization framework, the project drives advances in the technical and methodological level of road infrastructure monitoring.

Chapter 2 provides a summary of theoretical research and modern trends in detection of road damage and computer-vision inspection systems and intelligent transport infrastructure. It discusses the methods to date, the data sets researchers have utilized, and the methods of evaluation that have been utilized- point out their flaws. This pre-establishes the research level of my study and defines the gaps where the thesis will address.

Chapter 3 is a review of the related work and literature, where it reviews the development of object detection models and its application to infrastructure monitoring. It examines the RDD2022 data set and other essential reference points in the industry, ensuring that the solution that I am developing is deeply rooted in the recent research progress.

Chapter 4 describes the research questions and the problem format which is used throughout the project. It outlines the assumptions that we intend to test and measures that we will be using to determine the effectiveness of the system. Concisely, this part is the logical link between the literature review and the actual implementation.

Chapter 5 is dedicated to the realisation of the suggested methodology and the creation of the hybrid architecture. It presents the use of the dataset, preprocessing pipeline, experimental setup, the network architectures, and the metrics of reproducibility, but without dwelling on the details of the low-level code. This chapter contains the technical description that should be used in reproduction of the results.

Chapter 6 explains how the system is evaluated to check and prove that the system works based on a set of controlled experiments. It describes the numbers that demonstrate its performance and discuss the way the detection responds to various environments, discuss how it scaled and how it is solid in real-world applications. In essence, this section provides the concrete evidence which supports the central arguments in the thesis.

Chapter 7 is going to be the last chapter in the thesis, is going to conclude the entire thesis and give the recap of what I have discovered, the difficulties of the process, and what we will be able to do next. I address potential system adjustments, how it can be implemented in practice, and how my work on smart infrastructure affects the scope of the field, in general.

2 Theoretical Background

This chapter provides the reader with the background knowledge to understand technical concepts applied in the thesis.

2.1 Road Infrastructure Damage Detection

The reason why road infrastructure is so significant is that it ensures that people are safe on the road, supports economic activity and makes the life of cities more comfortable. Such objects as cracks, potholes, odd bumps might actually reduce ride comfort and increase the likelihood of an accident unless we notice them and repair them swiftly. In most cases, checking the roads would imply conducting manual surveys or taking out special vehicles which are costly, slow and difficult to cover large networks. As cities grow and roads deteriorate, the demand to have automated and scalable checks has increased rapidly. New sensors and data collection technologies allow us to collect large amounts of road images under real-life conditions. That is why computer vision is becoming the solution of choice to do automatic road damage analysis - reducing costs and enabling more frequent, large-scale inspections. Thus, the issue of road damage detection is a burning research question in the field of intelligent transportation system [13, 14].

Early approaches relied on handcrafted features and traditional image processing tricks to detect the damage on the road but they had difficulties in dealing with the various lighting, weather and surface peculiarities. This resulted in reduced robustness and systems that were difficult to deploy. Then there was the arrival of deep learning and in particular CNNs and they basically replaced many traditional approaches. It is possible now to learn features automatically using the raw pictures. The current detection systems can identify and tag various defect types simultaneously and it is essential when there is a complex combination of damages in a single scene. Big data and competition challenges assisted the community in establishing a standardized yardstick and accelerated progress. The state-of-the-art road damage detection systems today consist of deep-learning methods [15, 13].

In addition to the knowledge of the location of the damage, actual road maintenance requires actionable information to make intelligent decisions. Identifying defects alone is insufficient without prioritization. The managers need to consider defect type, its severity, and its location so that they can allocate resources effectively. This has given some interest in systems that combine perception and top-level reasoning and prioritization. Combining object detection with rule-based or expert process allows us to cluster hazards in terms of urgency. Hybrid systems are more interpretable and aligned with the reality of maintenance processes. Recent literature emphasizes the importance of explainable and transparent decision support in such arrangements [16]. Concisely, road infrastructure damage diagnosis can be approached most suitably by combined systems that integrate solid visual perception with explicit decision logic.

2.1.1 Definitions

To be clear and consistent, this thesis adheres to the following definitions concerning the road infrastructure damage detection and automated hazard assessment. These definitions

define a general language that I will use in the chapters. The current literature on intelligent transportation systems emphasizes the importance of standardizing a terminology in uniting perception and decision-support modules [17].

Road Damage: This refers to any superficial, visible wear on pavement that may affect the structural integrity of the pavement, unpleasant driving experience, or endanger the safety of traffic. Common ones include longitudinal cracks, transverse cracks, alligator patterns, and potholes, which are also commonly defined by datasets such as RDD2022 [15]. In the given work, the road damage is considered a visual pattern, which may be identified in RGB images. The emphasis on dataset-based research points out that a consistent taxonomy is central to cross-country generalization [18].

Object Detection: Object detection is a computer-vision problem that identifies and labels all the objects in an image simultaneously. In contrast to image classification, it provides bounding-box coordinates and class probabilities of each instance detected. This is possible in a single forward pass by modern one-stage detectors like YOLO, and allows real-time inference to be done in a single pass. The surveys indicate that the transition to anchor-free, multi-scale architectures increases the efficiency and strength of the architecture [19, 20, 21, 22]. Intersection over Union (IoU) and mean Average Precision (mAP) are typically used to gauge performance.

Hazard Prioritization: Hazard prioritization refers to the process of giving each identified road damage a severity/maintenance priority regarding a set of rules or a set of operational requirements. When we know what and where the defects lie, object detection informs us of the nature of the defects but prioritization converts this to maintenance planning. Prioritization is applied in this thesis through a rule based expert system which operates on the structured detection outputs. According to recent applied infrastructure research, it is important to connect perception outputs with maintenance workflows in order to do it. [23].

Expert System: An expert system is a decision-support system that uses a collection of fixed rules in working with organized input information to imitate expertise. Expert systems make infrastructure monitoring more transparent and interpretable since they explicitly model decision criteria, rather than making savings on black-box predictions alone [16]. The recent studies of hybrid AI in favor of modular integration between data-driven models and systems of symbolic reasoning suggest that they provide support to each other (and can be used interchangeably) [24].

Generalization: Generalization implies that a model can continue to perform well on the unseen information that was not included in the training set. This is particularly critical in road damage detection due to lighting variations, road material variations, camera angle and environment. It is important to evaluate on a held-out test set therefore to determine real-world strength. The domain robustness is pointed out as a requirement of work that should be present in object detection systems that are deployment-ready in the article of Hendrycks (2021) on the subject matter of robustness benchmarking. [25].

2.1.2 Types of Road Damage

Deterioration of the road surface manifests itself in a variety of ways, and each one of them is associated with its own group of reasons and problems with maintenance. The understanding of these categories of damage is the most important thing should we desire to construct a detection system, not just technically sound, but viable in the real life. Infrastructure-related studies have recently indicated that an automated system must be able to detect various types of defects rather than merely indicating a signal that something is wrong [26]. The various types of defects usually imply that we must make decisions regarding various strategy of repair, and the manner in which we will allocate resources.

Longitudinal Cracks: Longitudinal cracks are aligned with the direction of traffic and they tend to occur due to traffic overloads, poor joint work or material fatigue. Such cracks may begin to be thin and nearly invisible, particularly when the picture lacks contrast. According to some more recent deep-learning papers, it is difficult to see the thin structures of cracks since they are small relative to the entire picture resolution [27]. It is the reason why a good multi-scale feature extractor is needed to ensure accurate localization.

Transverse Cracks: Transverse cracks are visible across the direction of the traffic flow and transverse cracks often occur during swings in the temperature or shrinkage in concrete. They are more likely to be clear in shape as compared to longitudinal ones and still their width and severity does change significantly. When a transverse crack is oriented parallel to shadows or lane markings, it may cause a disruption in detection. Surveys that have clustered pavement issues indicate that the surrounding structure motivates the model to distinguish transverse crack between road markings and road markings only. [28].

Alligator Cracking: Alligator cracking, also termed as fatigue cracking, resembles a network of related cracks that take bizarre shapes in the form of polygons. This type of damage indicates that the pavement is becoming weak and it normally implies that the road is deteriorating to become very bad. It is so rich in texture that all-igator cracks actually enjoy a hierarchical feature look-at in neural nets of fatigue neural nets [29]. A multi-scale aggregation method particularly comes in handy to capture the small pieces of cracks as well as the entire pattern.

Potholes: The potholes are simply the tiny dents on the surface that have been caused by the structural failure and the breaking of the material. This is because unlike cracks, potholes are 3-D in nature, and they leave powerful texture and shadow indicators in photographs. Nonetheless, light and camera angle can transform them, and make them appear very different. The most recent research on detection asserts that it is simpler to identify potholes when you bring a bit of contextual reasoning and reliable bounding box forecasts to the table [30]. It is extremely important to the maintenance that potholes are properly spotted since they are directly linked to safety.

These types of defects are not only useful to be sorted accurately in the context of test numbers but also in terms of what to fix first. In actual road-maintenance processes every type of damage has a varying level of urgency and cost. This is why a multi-class detection system, which is able to segregate all these classes, is important - it will allow us to directly feed the output into a prioritization pipeline that is based on rules, as explained in this thesis.

2.1.3 Environmental and Data Acquisition Challenges

In the real world, damage detection systems on roads should be functioning in a wide variety of environmental and acquisition conditions. The shots of roads are normally captured by moving vehicles using normal cameras, something that introducing motion blur, perspective distortion, and variations in sharpness imbalance. The lights vary a lot due to time of day, shadows, weather and seasons. The surface material and the texture of the pavements vary in different places and this influences the appearance of cracks and potholes. And all these give a lot of domain variability, which makes the deep learning models difficult to generalise. Current research indicates that environmental variability is among the key obstacles to the scaling of vision-based road-inspection systems, as well as cross-domain ones, which have been identified as the problem of robustness to date, e.g., scalability and consistency between different areas in terms of their visual characteristics and, consequently, visibility conditions [31, 32]. Training models on a small range of environmental diversity tend to be worse when used in new geographic or climatic settings.

The other large problem is the data acquisition protocols, and camera set-ups discrepancies. A camera height, the angle of mount, focal length, and resolution differ and alter the size of the object and the way the scene is depicted in the photograph. Even minor flaws will only appear as a handful of pixels in certain systems and, thus, it is very difficult to detect them consistently. Also, the damage may be partially obscured by the vehicles, pedestrians, or shadows. Recent studies have indicated that robust models of road damage detection must be able to address the scale difference and partiality of visibility [33, 34]. Multi-scale feature extraction and anchor-free detection has been proposed to minimize such issues, although variability in acquisition is still a major concern that reduces reliability in the area.

Lastly, there is class imbalance, and noisy annotations, which further complicate environmental robustness. Types of damage such as potholes are very rare as compared to longitudinal cracks or transverse cracks. The uneven data distribution may alter the learning of the model and reduce the sensitivity of the model to defects that are difficult to occur but significant. Besides, large datasets that have inconsistent labelling create uncertainty which can propagate during model predictions. The existing literature emphasizes the need to have sound evaluation procedures and proper preprocessing to ensure that the data remains trustworthy [35, 36]. These environmental and acquisition issues are necessary that must be addressed to achieve scalable and reliable road inspection systems.

2.1.4 Class Imbalance and Small Defect Detection

In other words, the issue of class imbalance is serious in road damage datasets. Categories of defects appear many times more than others. To take an example, longitudinal cracks usually prevail in the data whereas potholes or other significant structural collapses are relatively isolated. Due to such an imbalance, training models will focus on the prevalent classes and fail to cover those defects that are safety-critical, yet relatively unusual. That imbalance confounds the level of confidence that the model has in its classification and its localisation capability of the defect in object detection. Recent articles demonstrate that such uneven distributions of classes have been a challenge in practical detection problems to date [37, 38]. Unless we apply correct evaluation procedures, the performance of minority classes may be lost under aggregate measures such as global mAP. That is why the analysis of the classes by classes is necessary in case of current infrastructure-related detection systems control.

It is further complicated by the ability to pick tiny objects. A number of cracks are only few pixels, particularly when the camera is positioned very high. The smaller pixels the network works with, the higher the number of false negatives. Finally, even the texture of the asphalt in the background can give a deception to the model, that an opening is merely a design. Multi-scale feature pyramid and improved feature aggregation are addressed by modern detectors to address this issue. [39, 40]. Nevertheless, input resolution in the sensitivity to small defects does actually require the depth of feature representations and our training setup. Therefore, controlled experiments are required when comparing various models.

Practically, class imbalance and tiny-defect detection have a direct impact on the maintenance prioritization. Missing a pothole that is very rare, but serious, can be far more dangerous than labelling a minor cracks. Then, metrics can not simply be interested in global accuracy and we need to investigate per-class robustness and the confidence of the detections. Recent works have indicated the significance of reliability-conscious evaluation in safety-critical infrastructure [41]. We can address the problem of imbalance and small-objects in a controlled experimental environment and make detection systems more practical in practice. This study analyzes these aspects with class-wise metric analysis and

prioritization rules in order, so that the model we choose is operational as well as merely looking good on average.

2.2 Existing Methods

Recent literature shows that road infrastructure damage detection has advanced rapidly with computer vision and deep learning. The first one was to design features by hand and feed them into the traditional machine learning frameworks like support vector or random forests. Such techniques used edge detectors, texture features or frequency-based features to identify cracks and rough areas. They perform well in the controlled lab environment, but their behavior is quite poor under other lighting conditions, weather, or dynamic road conditions, difficult to scale to large-scale applications in general - which limits scalability - just in time to deploy large numbers of them at once, anyway [42].

The convolutional neural networks (CNNs) significantly improved performance. CNNs are able to gain hierarchical features directly by looking at raw images without going through the manual feature engineering process. These earliest prototypes of deep learning largely followed the same patch-based classification loop: divide the image into tiles, classify each tile separately, and then reassemble the results. That increased detection accuracy, but localization precision was compromised, and the sliding-window assessment was expensive in terms of compute requirements, which were enormous [43].

This has made most of the research today shift to object-detection frameworks that are capable of locating and classifying concurrently. Two step detectors such as Faster R-CNN first produce region proposals and later refine them. It is very accurate but computationally expensive, which makes it inefficient in real-time or large-scale road inspection systems of high accuracy but high complexity, such as deep learning and neural networks. [44]. Owing to this, one-stage detectors are on the rise.

Single-stage detectors are desirable in that they achieve one forward step of detection, and offer a better tradeoff between accuracy and speed of inference. In practice, YOLO family has become the most popular option in recent studies of road damage. YOLO models are trained to work in real-time and have been proven to work well on various road damage databases. Recent versions YOLOv5, YOLOv7, and YOLOv8 fall into such variants as multi-scale feature fusion, anchor-free heads, and training schedules, which aid them to remain robust to scale change and very small defects [28, 21].

Some work papers also assess these YOLO-based models on large-scale datasets like RDD2020 and RDD2022. They claim to have competitive detection accuracy at low inference latency levels [15, 45]. In those researches, attention is typically paid to such metrics as precision, recall, and mean Average Precision (mAP). Although those numbers are important to benchmark with, they do not provide a clear image of how those outputs of detection contribute to downstream decision-making in the maintenance domain.

In addition to the raw scores of detection, increased focus is placed on robustness, reproducibility and feasibility of deployment. Other researchers attempt hyperparameter sweeps or exhaustive sweeps in order to obtain additional points. Nonetheless, such approaches usually reach a limit in terms of interpretability and may be difficult to replicate [46]. The trend of clean is, then, toward controlled experimental designs with standard configurations where we can attribute more transparently the variation in performance.

The primary weakness of most of the existing methods is that they consider road damage detection as a single perception task. Most pipelines are merely keen on locating defects although they fail to convert their findings to effective maintenance initiatives. The management of infrastructure actually entails prioritization and not raw detection. That criticism has prompted hybrid systems, which combine deep-learning perception with rule-based

or expert reasoning, and has presented greater interpretability and relevance in operation [47]. This thesis is based on the fact that the detection stage and the prioritization stage in a pipeline can be clearly separated and cohesion is maintained in reproducible pipelines.

2.3 Evaluation Metrics

When studying road damage detection systems, we observe that they can only be evaluated using standardized metrics of object detection. These indicators allow us to objectively compare various methods. The evaluations in computer vision research must reflect the classification performance and localization accuracy as it is not sufficient to identify the type of damage but localize its location. The recent benchmarking articles note that a single metric can easily deceive us, particularly in practice when the types of defects are diverse and the data are skewed in favor of certain categories of defects [48]. That is why the contemporary assessment procedures often combine a number of complementary measures to provide a balanced evaluation of the model performance.

Detection quality is checked by precision and recall as the measures of back-end. Precision is information that indicates the proportion of the instances of prediction that are accurate, which indicates the efficiency of the model in maintaining false positive to a minimum. Recall informs us about the proportion of the ground-truth objects that are detected indicating sensitivity of the model to actual defects. When it comes to road damage detection, these two issues are highly valued: an excessive number of false positives will lead to the waste of resources of unnecessary maintenance, whereas a low recall will result in hazardous features passing through unnoticed. According to recent works, precisionrecall analysis is particularly effective when the detection problem is skewed, with rare but vital defect classes and requires high detection accuracy. As pointed out in the recent research, precisionrecall analysis is particularly effective in skewed detection issues, where rare but important defect classes require high detection accuracy. [49]. At this reason, precision-recall curves are very popular to study the performance in relation to various confidence levels.

Intersection over Union (IoU) is a metric that is typically used to measure the localization accuracy of the detection, comparing the similarity between annotations in the ground truth and the predicted bounding boxes. IoU provides information on the success of a model to locate the identified objects which is important in downstream maintenance planning. The average precision (AP), which is the area under the precisionrecall curve on a specific class, is often bundled into detection performance. The average Precision (mAP) is an average of AP scores used in all classes and is used as a benchmark in object detection studies. In recent assessment efforts, there are calls to report performance at multiple thresholds of the IoU to evaluate how performance robustness under a more stringent localization criterion is tested [50].

In addition to the more traditional accuracy-oriented measures of the performance, recent studies put a call to frameworks of evaluation that take into account the actual deployment limitations and relevance. Measures ought to be operational needs based, like our tolerance to false positives, the computational cost of the model and its scalability. Increasingly applied computer vision research claims that benchmarking must be consistent with real-world application contexts and not only with leaderboard results. [51]. This viewpoint is particularly applicable to infrastructure surveillance, where the results of detection must be able to make actionable decisions rather than merely indicating the raw accuracy.

Lastly, the recent evaluation studies also point to the necessity to move beyond the single-number summaries to deconstructing complex detection systems. These complementary studies, such as class-by-class performance breakdown and inspection of error

patterns provide further understanding of the strengths and weaknesses of a model. These multi-dimensional evaluation systems contribute to transparency and assist in informed system design decisions [52]. When it comes to the task of road damage detection, we can confidently evaluate the behavior of the model of this task in different defects and under different conditions of work through the application of these practices.

2.3.1 Precision and Recall

Precision and recall are the fundamental parameters of any detection test, and they are what we will be using to measure the performance of our road-damage recognizer. Precision informs us about how many of the detections we have indicated as correct are in fact correct and hence high precision translates to lower level of false alarms which is important when maintenance crews are called out. Recall, in contrast, measures the number of the actual defects that we were able to detect; failing to detect a crack or a pothole can be a life threatening event. Practically, measurements of road-damage have a very skewed distribution, and we often consider the two measures simultaneously and not individually [48, 49].

2.3.2 Intersection over Union

The preferred measure of checking the accuracy of localising the damage is intersection over Union (IoU). It compares ground-truth box with our predicted bounding box. We view a detection as correct when the IoU surpasses a threshold, which is usually 0.5 in road-damage criterion. This makes sure that a defect is not merely detected, but also located precisely, which is critical in designing repairs. The vast majority of recent research adheres to this IoU standard to enable comparing the results of their study directly with other studies in the field. [15, 45].

2.3.3 Average Precision and Mean Average Precision

This is because Average Precision (AP) allows us to calculate the combination of precision and recall at all levels of confidence to provide a more comprehensive view of our detector. It proves particularly effective when dealing with skewed data sets and the confidence score is of interest. With a number of classes of damage, we compute an AP, and average the results to obtain the mean Average Precision (mAP). mAP now becomes the usual metric of road-damage competitions since it weighs localisation precision with classification precision. The trade-off that we seek is evident in many papers that use the mAP@0.5 metric, which clearly demonstrate the trade-off that we are seeking to achieve in the future [13, 14].

2.3.4 Metric Selection for Road Maintenance Applications

Though the conventional measures give good scores, we must remember those who are going to be using our system. In a maintenance pipeline, high recall implies that there is a minimal number of dangerous defects that get through it, whereas high precision helps to ensure that false alarms are low. This is why the current applied work emphasizes reporting a number of complementary metrics, as an alternative to single-number reporting. This multi-metric method can assist decision makers to evaluate the extent to which a model is indeed appropriate to be used in the real world of deployment [16]. We will adhere to the canonical metrics of object-detection in this chapter to be similar to the literature, yet we also consider practicality to prioritize repairs.

2.4 Research Gaps and Limitations of Existing Approaches

Although much has been done in automated road-damage detection, there are some major issues that remain. The majority of the papers are pushing the boundaries of the precision, the recall, and mAP without considering the impact of these numbers on the maintenance workflows themselves and how to interpret them as necessary or beneficial in the first place [13, 14]. When researchers focus on the best score on a benchmark, we find ourselves with too complicated models and hyper-parameter optimization that is difficult to reproduce or execute once in the field. The outcome is the discongruity between what appears good on paper and what a traffic authority can practically utilize.

The other weakness is that detection is considered to be a separate perception task. Existing systems will produce the boxes and labels, but not inform us how to prioritise the relief work [16, 47]. A visual cue is not enough, a decision maker should be presented with severity, context, and clear explanation. This gap can be addressed by rule-based reasoning or an explainability layer.

Lastly, there is the issue of reproducibility. Various researchers employ either tailored splits or unofficial preprocessing methods, so it is unlikely that this would be reliable when compared to different researchers [53, 54]. Although there are datasets such as RDD2022 offering a standardized benchmark, discrepancies with the way the models are trained are still revealed. We require a pipeline that records all the tweaks and holds the experimental setup as controlled as we can so we can increase our trust in any gains we will declare. This thesis pursues the philosophy by distinguishing between perception and decision-making and putting emphasis on transparency and practical implementation.

3 Related Work

This chapter is a review of all the literature that is related to automated road damage detection, object detection frameworks, and hybrid AI-based decision systems. The argumentation is divided into parts that reflect the main elements of the methodology I am offering: deep learning-based perception, evaluation strategies, and hybrid decision-making systems. Within every section, I have put together studies that are similar in terms of methodology and indicated the current developments that align with this work.

The summary of the literature is shown in Table 3.1.

Table 3.1: Summary of related work grouped by subcomponent and approach.

Subcomponent	**Approach family**	**Representative papers**	**Data**	**Key take-away**
Detection	One-stage detectors (YOLO)	[28, 21, 15]	RDD2022, others	Strong speed vs accuracy trade-off
Detection	Two-stage detectors	[44, 55]	Mixed	Higher accuracy but heavier inference
Preprocessing	Augmentation and QC	[56, 57]	Mixed	Improves robustness if controlled
Decision support	Hybrid, explainable prioritization	[47, 58, 59, 60]	Mixed	Converts detections into actionable priorities
Evaluation	Metrics and protocols	[48, 61, 53, 54]	Benchmarks	Standardized reporting improves comparability

3.1 Deep Learning-Based Road Damage Detection

Recent literature shows that deep learning has become the predominant method to use in automatically identifying road damage since it is capable of learning rich and hierarchical features directly out of raw images. The recent articles continue to demonstrate that convolutional neural networks (CNNs) are superior to human-designed feature algorithms under varying lighting and road-surface conditions [62, 63]. CNNs are capable of processing any shape of crack and tricky backgrounds, unlike popular edge-based or texture-based tricks, without requiring carefully crafted features that foreseeably succeed or fail, which is a significant benefit when it comes to large-scale inspections where conditions cannot

be carefully controlled. It has also been accelerated by the open access of the datasets such as RDD2022 that have allowed to test the models in a standardized manner. Today, the emphasis put by researchers on robustness and cross-domain generalization has been more developed compared to pursuing higher benchmark figures, and this is what is being required in real infrastructure management. Subsequently, the deep-learning road-damage detection has matured beyond proof-of-concept but can be used in practice.

The object-detection models are generally divided into two camps: the two-stage and one-stage models. Two-stage detectors, including Faster R-CNN and its modifications, generate region proposals and then classify them and optimize bounding boxes. These models tend to achieve high accuracy, but are very expensive in computation and latency, negatively impacting scalability in real-time inspection pipelines [64]. Inference speed is a significant bottleneck in activities that pass a significant amount of images through the system. Several comparative studies indicate that despite the fact that two-stage detectors can cause the localization accuracy to be displaced a bit higher, the trade-offs are not normally justified in large deployments [65]. Therefore, much more recent practical work is towards architectures that maintain the solid detection performance and reduce inference cost. This accuracy-versus-efficiency tradeoff directly determines the choice of models used in production systems.

That is why one-stage detectors have gained popularity in the recent past. YOLO-style models are particularly popular as they are end-to-end and can be implemented in real time. Recent research demonstrates that YOLO-based networks are able to achieve comparable competitive mean Average Precision scores with significantly lower inference latency compared to proposal-based structures [66, 32]. These models have been made more sensitive to minute cracks and road patches by improvements in feature pyramids, anchor-free heads and enhancement of training strategies. These are especially applicable in the case of pavement inspection where defect sizes may range wildly. RDD2020 and RDD2022 experiments prove that reduced-weight YOLO variants can generalize well even to other geographic regions of interest. [67]. Therefore, one-stage structures are a good option in scalable road-monitoring implementations.

Despite all this, there are still a couple of challenges. Most research focuses on increasing mAP without considering the impact of detection outputs on the maintenance-decision processes. Other benchmark-inspired methods go even deeper into hyper-parameter optimization and extensive augmentation which may adversely affect reproducibility [68]. Domain shift is another obstacle because models that are trained on a particular area might fail when applied in other regions [69]. The concerns emphasize the necessity to plan the experiments and evaluation protocols properly. This thesis addresses these gaps by prioritizing reproducibility, establishing controlled configuration comparisons, and distinguishing perception from prioritization steps. The proposed pipeline links detection to decision logic aligns the performance of the model with real-world infrastructure-management requirements instead of merely trying to achieve benchmark scores.

3.2 Object Detection Architectures and Design Trade-offs

The architectures of object detection have considerably changed throughout the last decade as a result of the necessity of the balance between the localization precision, the classification accuracy, and the efficiency of computation. The current detectors can be broadly divided into two-stage and one-stage systems, each having its own benefits and limitations [19, 20]. Two-stage detectors use the precision of the localization by refining proposals whereas one-stage detectors use unified prediction pipelines which are optimized to be fast. This trade-off has a direct contribution to the scaling of the system and its deploy-

ability in the case of infrastructure inspection tasks, e.g. road damage detection. Models that are too complex and expensive in terms of computation might not work well with large-scale monitoring systems involving the processing of thousands of images per day. On the contrary, lightweight models should not lose high sensitivity to minor defects to high robustness. Thus, the design choices of architecture should be considered not only regarding the benchmark measures but the operational ones. This balance characterizes the direction of architecture taken in the current study.

According to the recent comparative research, the multi-scale feature fusion processes can be used to detect small and irregular objects, such as fine-grained cracks and narrow surface discontinuities, more effectively than without such mechanisms in place and they can be effective even when the objects are very small. [70, 71]. The designs based on the feature pyramid maximize the contextual understanding through aggregation of information on varying spatial resolutions. These constructions enable the detectors to be sensitive to smaller texture changes, with the global structural information still available. This property is necessary in the pavement inspection cases where the size of objects differs significantly. The empirical analysis demonstrates that detectors that lack effective multi-scale representation have lower recall of small defects [72]. As a result, architectural designs that focus on feature pyramid networks and cross-scale information communication have hitherto become commonplace in contemporary detection pipelines. These results justify the architectural decisions as outlined in the implementation chapter.

The other important design issue is in relation to anchor-based and anchor-free detection strategies. Anchor-based algorithms are based on fixed bounding box priors, and have to be carefully tuned to dataset properties. Contrary to this, anchor-free methods forecast object locations and regression biases explicitly, both of which are simpler to configure and more adaptable to changing conditions [73, 74]. In the case of heterogeneous datasets like in the case of RDD2022, anchor-free designs minimize the heuristic modifications required of the dataset. Recent studies illustrate that anchor-free detectors can be competitive or even outperform anchor-based approaches while using fewer hyperparameters [75]. This quality is in line with the reproducibility goals of the current work. Reduction of manual tuning minimizes experimental bias and increases transparency of the methodology. Thus, principles of the design of anchors were taken into account in the architectural choice.

In addition to building design, computational performance and scalability are becoming more of a key focus in practical object detection studies. The models can be deployed on edge devices and systems with resource constraints with lightweight backbone networks and efficient training pipelines [76, 77]. Hardware availability and energy consumption should be taken into consideration in large-scale infrastructure monitoring environments in addition to detection performance. Research highlights the importance of marginal gains in mAP not necessarily indicating significant computational demand increases [78]. This view has resulted in an increasing focus on a balanced model selection approach that incorporates accuracy, latency and robustness indicators. The current research paper adheres to this direction of applied research by comparing the model variants in controlled experimental conditions and choosing the configurations by the empirical trade-off analysis. This kind of approach is in such a way that architectural choices can be made in line with the feasibility of real world deployment and not only in line with benchmark optimization.

3.3 Hybrid AI and Explainable Decision Systems

Recent studies highlight that accurate detectors can be learned using deep-learning models, and it is difficult to understand what makes those layers specifically pick something because there are too many layers. In practical infrastructure projects, stakeholders require a clear

explanation as to why one piece of road is prioritized above the other. That is why some recent studies have been paying a lot of attention to the hybrid AI systems that involve the use of both data-driven models and rule-based or symbolic reasoning modules. The concept is to maintain predictive ability of the neural nets and to make the reasoning chain transparent to make it easy to audit and justify. [47, 79]. I read through and realized that these architectures are gaining ground particularly in safety-critical applications such as infrastructure monitoring.

The thing that is going on this year is that people are actually combining deep neural nets on the perception tasks, and then are piping their outputs into a structured decision engine that represents domain knowledge and constraints. The separation allows you to specify the prioritization logic separately, adjust it, and test it, without adjusting the network weights. Articles on industrial inspection and predictive maintenance indicate that such a combination in fact enhances transparency. [80]. It looks highly applicable to road maintenance since you are normally compelled to rank choices depending on established safety standards and not only security rating of a model.

XAI studies, contrary, are attempting to apply techniques such as saliency maps, feature attribution and other post-hoc explanations in order to disaggregate individual predictions. It is cool and these local explanations are not the entire logic of decision in the entire system. [81]. The hybrid framework that I have discovered in the literature underlines the fact that detection and prioritisation are two different layers and that simplifies the logic of the system and puts it in an easier place to control in case of the infrastructure of a city.

The recent literature also exhibited a tendency that I noticed, which emphasizes explainability and reproducibility in the same direction. [82]. Complicated ensembles may squeeze too little out of performance, and frequently clarity and deployability may also suffer. This is why most of the projects are now preferring modular designs that can allow tracing the performance gain to a definite component and make the decision criteria crystal clear. My personal thesis is a continuation of this trend, in which perception and prioritisation are cut and interoperated into distinct blocks. Hopefully, the student friendly design will make the system readable and retain its own against competitive benchmark numbers.

3.4 Evaluation and Benchmarking Strategies

Credible assessment is extremely significant in the area of object detection research since you wish to ensure that the numbers you observe are the actual representation of how well the model will perform in the reality, and not solely on a particular dataset. Recent work emphasizes benchmark-driven experiments are actually the in-thing as it can be used to compare across architectures and training approaches in a standardised manner [48, 13, 51]. Large datasets containing pre-defined splits provide us with a controlled environment to test performance, and that contributes to reproducibility and reasonable comparison. Naturally, we must draft the evaluation protocols very thoroughly lest we come up with an optimistic bias. The separation of the training and validation and test sets is an absolute requirement. This is particularly applicable when we are dealing with an applied area such as infrastructure inspection and hence the design of the assessment directly influences the degree of believability of the results.

Recent work emphasizes not only metrics but also clear experimental protocols [53]. Fixed splits of data, preprocessing consistency, and controlled hyperparameter settings are required in case you want a reproducible benchmark. The detection metrics can be significantly disproportionate by even minor modifications to augmentation strategies or threshold settings. This is why the standardized evaluation pipelines are being stampeded on any fresh paper in the field of applied AI research. You should not hide details (including

hyperparameters, dataset splits and thresholds) when you report them: it will allow other people to get the same results. In the case of AI systems involving infrastructure, verifiable and auditable validation is not a pleasant addition, but rather a necessity. None of this methodological rigor increases the reliability of the experimental conclusions any more than how could you not see the tight linkage between benchmarking strategy and the reproducibility?

In addition to the common aggregate statistics such as mean Average Precision, recent studies also explore class-wise and threshold-based analysis of classes and threshold-based analysis [14]. Detection systems with objects can do actually different things with their categories, particularly when the data is skewed. The type of defects may be super rare in a road-damage scenario, but they may be super important to safety. When you separate the performance by class, you will see a far clearer picture of what strengths and weaknesses of the system are. The inclusion of a multi-threshold assessment, i. e. measuring with relaxed and strict IoU requirements, allows you to visualize the strength of the localization itself. It is that level of detail, which makes us smarter in architectural decisions and understand when to use a model.

And in practical research, you can not simply think about accuracy. The inference time, memory consumption and scalability are massive deal-makers in real-world road-inspection configurations [51, 78]. Speed-accuracy trade-off is the reason why the modern benchmarking frameworks revolve around it. Being a lightweight model may fail to preserve a small amount of accuracy but will gain a lot on compute, which can be a game-changer in the area. On the other hand, a heavier model may only present marginal benefits to a colossal asset strike. A balanced approach that considers predictive performance and operational constraints is therefore necessary. Such multidimensional perspective perfectly fits the interests of infrastructure monitoring teams in the real practice.

3.5 Deployment and Practical Constraints in Vision-Based Systems

In deployment settings, the issue of detection accuracy is a big concern in theory, but in actuality, when you need to roll out a system, pocket-real-world stuff presents a lot of twists. The infrastructure surveillance that uses vision will have to withstand every kind of wild environment: day/night lighting variations, unstable weather conditions, shaky camera as a car passes a vehicle, and cameras that do not always have the optimal location to watch the infrastructure [83]. The model loses a lot of points on these "real-world" variables that you never had much attention on in pretty clean benchmarks. That is why many researchers today are straining to do robustness tests which are modeling the system behavior when under attack. As an example, speed on a road, the number of pixels in the camera, the vibrations on the road, and so on can all confound the locations of the cracks and the consistency of the system in marking them. When we fail to inject these variable conditions into the equation we are just making guesses of how the system will behave in the trenches. Making deployment a success is not necessarily the top score on an abstract test, it is how the system can survive in real world conditions.

Another giant is the technology it is using. The roads are extensive, pictures build up at a rapid rate and we require results almost immediately or at least not later than the next frame on a thin edge device not a server farm. The literature is replete with slick ones-TinyDet, MobileNetV3, and other lightweight designs that are geared towards an optimal bite of accuracy and speed only [84]. The failure of an over-engineered net to fit on an edge GPU or consume a ton of battery is not a deal. Hence the reason why the fraternity scows off ever-slender models to achieve knowledge distillation, pruning and quantization

in a bid to succeed in the speed accuracy war. However, one must test the punch-line of it all you see; you do not want to put the health of the model at risk just to run a little faster.

Moreover, it becomes problematic with scaling. When a city wishes to roll a monitoring system over the entire country, the monitoring system must merely follow all those additional images - without having to retrain itself every time the road is altered, or the cloud shifts. Researchers are developing tricks of continual learning that enable a network to update its weights, refresh its knowledge base and continue to roll as the input distribution changes across various regions [85]. This implies that a city government does not have to stamp an entirely new model on a seasonal basis. Modularity is also provided; the crack-detector can be updated independently of the prioritizing engine without dropping the entire stack. These detachable examinations render the instrument deployment-ready and simpler to keeping up.

Lastly, it is unlikely that a deployment that cannot justify its decisions will pass. Within the government community, you must have a clear line of reasoning in showing why a certain maintenance team has been assigned to a section of highway. Black-box networks with high scores cannot explain why but still work, which is not only dangerous but also a regulatory nightmare to the regulators themselves [47]. The fashions of hybrid ideas, therefore, a deep net to see crack shapes and a a rule-based module that applies thresholds based on severity indicators defined by the agency. Coupling perception and decision-making will leave the stakeholders comfortable, as they are in a position to observe the stream of information and explore consequences. The entire success of a system, therefore, is not as much about raw measures, but is more about describing your work and fulfilling the requirements of governance.

3.6 Summary of Literature and Research Gap

The literature shows that the automated road damage detection has made significant advancements courtesy of deep learning-based object detection systems. Large-scale benchmarks, particularly those based on YOLO, are seeing current one-stage detectors perform quite well on RDD2020 and RDD2022. The advancements in multi-scale feature extraction, intelligent training approaches, and model scaling have indeed assisted the systems to remain strong in various environment conditions [15, 14]. And to add this to it, the recent research is emphasising the significance of the standardised evaluation, reproducibility, and deployment feasibility as the significant elements of applied computer vision systems [53]. All this has provided a technical base to the vision based road inspection, though the majority of the current work is still primarily concerned with the quality of performance the models achieve on benchmarks. Converting those results on detection into structured maintenance is still relatively unexplored, and that disconnect between perception research and actual infrastructure maintenance is quite clearly visible.

A common drawback is that the concept of road damage detection is considered as a single perception problem. There are numerous papers that maximize the mean Average Precision without considering the way those detection results are being incorporated into operational prioritisation. Managers must have the ordered or categorised output as opposed to the raw bounding boxes in real road maintenance scenarios. Since no organised prioritisation mechanism exists, purely detecting based systems are not able to make a huge practical difference. Although explainability and hybrid AI methods are beginning to gain traction, they have only recently appeared in road damage detection pipelines, albeit in a much less frequent occurrence than other components of the pipeline [47, 16]. This implies the need to have architectures which explicitly decouple perception and decision logic - in

that case we obtain transparency but do not lose detection accuracy. To solve the problem, a hybrid system of solid object detection and interpretable rule-based prioritisation layer would be used.

The other problem is that it does not have controlled experimental comparison in reproducible conditions. Although top models are benchmarked, it is not common to systematically test different controlled configurations. Such factors as input resolution, training time, model size can change results but are hardly transparently reported. The concept of reproducibility has been identified as a larger issue when it comes to machine learning research [53]. Infrastructure decision-making, the design of reproducible experiments is important since the decisions made regarding deployment can be based on the purported performance difference. Ensuring that we split the data consistently, use the same hyperparameters and evaluation protocols increases reliability. A clear reporting also aids the future studies in making fair comparisons. Therefore addressing reproducibility is not only a scientific nicety, but it is useful as well.

Considering these shortcomings, this thesis suggests a well-organised and repeatable pipeline consisting of the combination of deep learning-based detection and a rule-of-thumb expert system prioritisation. I will appraise the detection stage in controlled experimental conditions to systematically examine performance trade-offs. The prioritisation stage will then convert detection outputs into operational categories of maintenance to enhance operational relevance. This separation of perception and decision logic will make the system easier to interpret and apply in a more modular manner. This is what the infrastructure managers actually require and it also offers an experimental basis on which others can reproduce. Therefore, the thesis does not only make a contribution to the benchmarking of the detection, but it also relates the perception to the decision support in road infrastructure monitoring.

4 System Concept and Design

This chapter extends the context of the problem presented in Chapter 1 and specifies it into a system design and problem formulation.

4.1 Problem Abstraction and Design Rationale

Road infrastructure surveillance is a large-scale real-world problem affected by varying illumination, adverse weather, camera viewpoints, heterogeneous pavement materials, and diverse damage types. The manual inspection of roads is challenging, too costly, too slow and the human factor comes into play. In practice, this can be framed as a vision problem: identify, find, and categorize road damages in changing conditions. Any automated system must balance detection accuracy, robustness, and scalability, while being evaluated under repeatable protocols to avoid results caused by data preparation issues or training bias [86, 87].

This thesis is based on the premise that it is possible to detect, with certain confidence, the existence of road surface defects within the image of one of the single cameras, recorded in the chaotic, real environment. Compared to the clean industrial laboratory scenes in tutorials, road pictures are full of intra-class variation and class disparity. Certain flaws occur significantly less frequently than others do [88, 89]. That is why this motivates the use of deep-learning object detectors which could acquire the hierarchical features with direct data learning [19, 20]. However, pure data-driven models do not usually convert raw boxes into the maintenance advice in the real world. Road officers request a sense of urgency, but not coordinates. The issue is not merely detection, though, it is how to transform what the model detects into prioritized actionable information. The system must be able to communicate between perception and decision support.

To accomplish that, this thesis integrates a deep-learning perception block and a rule-based reasoning layer. The task of the detector is to identify the location of the damage in the image and the expert system then compares the detection data with domain rules that are designed by hand. The division of the work simplifies it and ensures increased clarity, transparency, and maintenance ease of working on it [16]. It also allows the incorporation of knowledge that cannot be easily gained by data alone, such as how to balance the severity of damages, or how to balance several cracks. The rule layer provides me with a fixed interface on sensitivity checks, since I can adjust the thresholds without retraining the model. It is a practical advantage when policies are more advanced than the world model. This hybrid arrangement has the overall advantage of enhancing auditability and aligning the pipeline with the real decision requirements.

A modular pipeline is adopted instead of a monolithic end-to-end design. The decision is reminiscent of what we had heard in project seminars, flexibility in experimentation, reproducibility of results, and the ability to extend it later on in life [53]. Each of the modules may be tested, adjusted or replaced without the entire stack being lost. The division of the problem into data acquisition, visual detection, post-processing, and expert evaluation steps is reflective of best practices in the design of an applied CV system. In addition, a modular system enhances traceability, since intermediate products can be stored and checked on a run-to-run basis. This is because the same detection outputs can

be used to test various prioritization policies, eliminating the confounding factors when I analyse system behaviour. This argument is the basis of the architectural decisions that I will elaborate in the following chapters.

4.2 Research Objectives and Questions

In essence, the proposed project is concerning the construction of a reproducible and interpretable road hazard spotting and prioritisation pipeline based on deep learning and rule-based reasoning. Rather than simply paying attention to the accuracy of detection, as most papers do, I will be breaking down the perception process and the decision process in order to have a more transparent system, easier to implement and actually viable in the context of real world infrastructure management. Here, it is focused on the general utility of the system, not only on the beating of benchmarks. The reproducibility is of high priority that anybody can replicate and extend my findings. The deterministic expert layer and clear rules of post-processing provide interpretability. The practical component is concerned with transforming raw detections into useful maintenance priority labels. On the basis of these objectives, I have prepared the following research questions, which combined, form what I would be measuring and demonstrating to become a success of the pipeline.

- **RQ1:** To what extent does a lightweight object detection model with the use of the YOLO architecture address the RDD2022 dataset in controlled experimental settings, considering the metrics of precision, recall, and mAP?
- **RQ2:** How does the trade-off between the accuracy of model detection and its computational efficiency compare between light-weight and heavier model configurations?
- **RQ3:** What can I do so that I can transform identified road defects systematically into tangible maintenance priorities through a well-defined, rule-based expert system?

These research questions guide both the design choices, the experimental model and the assessment methodology that I will discuss later in this thesis. I will be putting them through close experiments, the standard detection measures, and a close analysis of the post-processing of the expert system. Ultimately, they assist in ensuring that I am considering not only the individual elements but also the actual value of the system in the real world in terms of maintenance planning.

4.3 Problem Formulation

Consider an input road-surface RGB image $I \in \mathbb{R}^{H \times W \times 3}$. The perception module outputs a set of detections in the form:

$$D = \{(b_i, c_i, s_i)\}_{i=1}^{N},$$

where b_i are the bounding-box coordinates, c_i the defect class, and s_i the confidence score. That is, the detection step converts the raw pixels into a clean list which could be assessed in a quantitative manner. We usually evaluate it using matching on the basis of IoU, class-wise accuracy and recall, and macro-averaged measures such as mAP. The set D is purposely model-agnostic such that any detector can be plugged in. More to the point, the raw output does not encode what maintenance to take place next, regardless of whether the detector is super-accurate. Users in real deployments require actionable summaries rather than raw detections and the speed with which they have to respond. It is the reason why

we require an additional decision support layer that will convert detections into action plans.

The perception output is then fed into the decision-support task which then assigns maintenance priorities:

$$P = f(D),$$

where $f(\cdot)$ is a rule-based function that classifies each detection into low, medium, or high priority, depending on the type of defect, the confidence of the detection, and the thresholds set by the user. The role is able to weigh various classes, impose minimum confidence gates, and merge details of numerous detections. In contrast to an educated neural model, the prioritization logic is non-probabilistic and can be inspected or edited without retraining the perception stage. It is convenient when the policies vary or the constraints can vary among regions. Detection quality and decision logic can be independently tested since they are different and perception and prioritization are distinct. It also allows us to perform controlled ablations - solve only f( ) with D constant. Collectively, this architecture creates an open channel linking computer-vision results with decision support information.

4.4 System Architecture

The proposed system is a modular computer-vision pipeline for automated road-hazard detection and rule-based prioritization. The architecture does not try to implement a single and large end-to-end solution but rather divides things into perception, decision making and evaluation. In that manner we can observe precisely what each component is doing, replicate it and modify each module separately. Applied computer vision uses modular designs and is most often used in safety-critical applications, such as infrastructure. [5, 36]. The entire pipeline is a modern ML best practice, transparent flow, and controlled data routing, in order to reduce hidden dependencies. It is also a way of taking clean ablation experiments, as well as comparing model setups systematically.

At a high level, the system consists of four main steps: data ingestion and preprocessing, visual hazard detection, post-processing/ aggregation, and expert-system-based prioritization. In the first place, raw road images undergo standardization through a preprocessing step that corrects the resolution, verifies the quality of the annotations, and correctly divides the dataset. Then, processed images are fed into a deep-learning object detector that detects and locates road surface defects. These are filtered and stitched to produce structured data, which is now ready to make downstream decisions. This simulated environment is consistent with perception-based system design of decision systems in intelligent transportation and infrastructure surveillance systems [90, 91]. It is important to separate detection and prioritization since the change in policy may occur even in the case when the detection model remains unchanged.

4.5 Overall System Pipeline

This section describes the system pipeline adopted in this work. Figure 4.1 illustrates the entire process, beginning with curating the data and training the models to the evaluation of the output and providing the decision-support. It is a definite division between the learning and the reasoning based perception. Such separation is quite useful since it can reproduce everything: any intermediate products could be stored and used in further experiments. Besides, it simplifies the process of decision making, which can be easily audited.

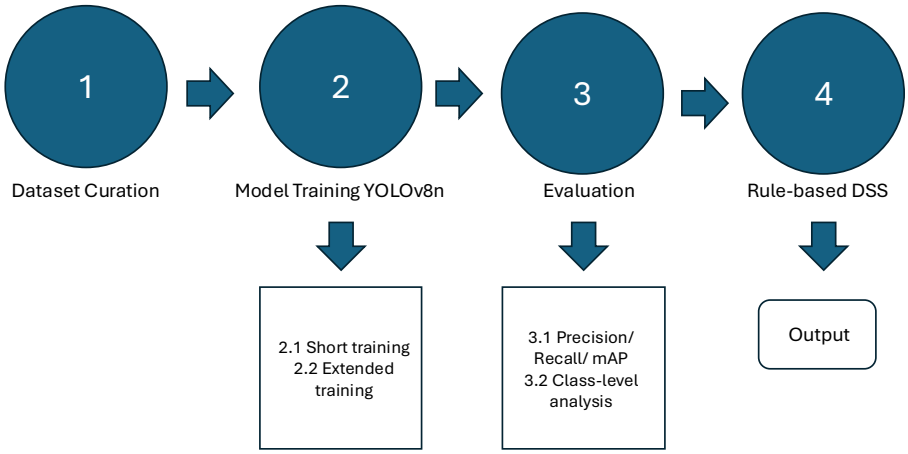

Figure 4.1: Overview of the proposed pipeline, dataset curation, model training, evaluation, and decision support by rules.

4.6 Detection Output to Prioritization Interface

The hybrid system can only work properly under the conditions that the connection between perception and reasoning is evident and consistent. I intentionally held that connection close to the organized characteristics of detection that are easy to analyze and verify in this undertaking. The detector provides us with bounding boxes, class labels, and confidence scores, and that is definitely sufficient to construct a policy of auditing prioritization. This interface is mapped on the priority categories as indicated in Figure 4.2. This allows us to avoid reliance on hidden features or raw image content, which otherwise would cloud the waters by relying on the structured detection outputs. It further simplifies evaluation, since we can reuse the same set of detections and simply rescore them with a new set of rules. There is also traceability, which is supported in the interface: any priority label may directly refer to concrete detection evidence. These design options are in line with the explainable and interpretable AI principles [47, 79]. In brief, decision logic will remain tweakable, testable and aligned with operational constraints through the interface design.

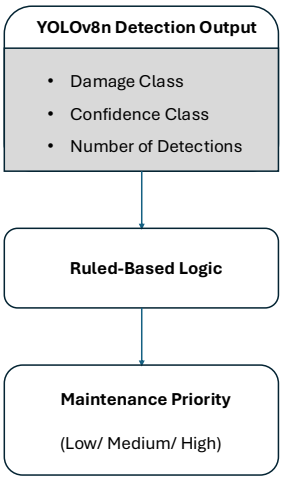

Figure 4.2: Interface between YOLOv8 detection outputs and rule-based prioritization logic.

4.7 Design Trade-offs and Justification

There were several distinct trade-offs between performance, efficiency, interpretability, and reproducibility that I made in this project. By selecting a lightweight detection model, we can perform fast inference over a large image collection and may even operate in older hardware that may be available in resource-constrained settings. I also separated the system into distinct modules to be able to test the parts of perception and decision separately and to be able to replicate everything. Incorporation of a solid post-processing step to maintain the outputs constant and reduce the number of false positives through the application of constant thresholding and non-maximum suppression. Deterministic expert layer provides a straightforward, auditable prioritization rule which can be adjusted to alternative maintenance policies. Lastly, I divided component-level metrics and system behaviour in order to avoid confusion between pure quality in detection and policy choices. These decisions align with the reproducing ML recommendations and the production ready system design principles in ML systems [53, 51]. Table 4.1 recaps the key architectural choices and their significance.

Table 4.1: Key design trade-offs considered in the proposed system.

Design aspect	**Choice**	**Rationale**
Detection model	YOLOv8 lightweight configuration	Enables scalable inference with competitive performance
Pipeline structure	Modular staged pipeline	Improves reproducibility, ablation capability, and traceability
Post-processing	Thresholding and NMS	Stabilizes predictions and reduces spurious detections
Prioritization	Rule-based expert system	Provides transparency, auditability, and policy flexibility
Evaluation focus	Metrics plus class-level analysis	Captures imbalance effects and supports operational interpretation

4.8 Data Flow and Component Interaction

This section describes the data flow of the system and how the respective parts of the system interact when training, inference and during evaluation. A crystal-clear data pipeline is exceedingly significant in maintaining consistency, tracking and reproducibility of our experiments. Applied ML systems often result in fuzzy data dependencies as the massive source of bias and you simply cannot reproduce the results [92]. That is why the system imposes clear-cut transitions between processing stages, each of which has clear inputs and outputs. We begin by retrieving raw images of the road-damage dataset and an annotation/validation step is executed on the annotations and the images. The pre-processed results are ensured to be immutable and kept in isolation to ensure that nothing spills over between train/val/test splits [92, 54]. The processed datasets are the only input to the visual detection module, which maintains the data prep in a sandbox of its own as the model learns. After preprocessing, the fixed train/validation/test splits are used as inputs to the detection module. During inference, images are processed without modifying the dataset splits or labels, ensuring consistency.

The detector outputs structured predictions including bounding boxes, class IDs, and confidence scores. We do not feed that raw directly to the evaluation or a decision on the final outcome, we feed it first to a post-processing layer. That layer performs thresholding, spatial filtering and possibly fuses detections between images when required. You can imagine that it is a defined interface that allows the perception layer to communicate with the reasoning layer, and to reduce coupling between modules. [4, 93]. The output of the post-processing is that of the detection, which then is inputted to an expert-system that uses a series of deterministic rules to determine the priority of maintenance. The expert system does not handle raw pictures, but only structured outputs hence we remain modular and interpretable. Each expert layer decision can be linked to specific detections and rule triggers, which is very convenient in the infrastructure-related decision support audit trails [47, 79].

In the process of evaluation detection metrics are computed prior to the expert system coming into effect, so that we can observe the true performance of the model without the decision logic obscuring the metrics. Next we mix the expert outputs and determine the uniformity of the prioritization and whether they are useful to the team. [79, 53, 54]. Concisely, this clear flow of data and strict control of the components provide us with an excellent foundation of reproducible experiments and practical application.

5 Methods

5.1 Research Methods

In this work we are, therefore, approaching the entire automated road hazard detection issue with the use of computer vision. In essence, we will construct a deep-learning pipeline and what we will do is to put this into a rule-based system to determine what is the priority. We are applying quantitative procedures to measure literally how well we are detecting road damage instances in the same ways that people in the field do. This setup enables fair comparisons of model variants under controlled conditions. The methodology is designed to support practical road-inspection scenarios and realistic operating conditions.

The methodology is divided into two stages: perception and decision-making. First, a CNN-based detector is trained to localize and classify road damage types. Multiple configurations are evaluated under controlled conditions, and the final configuration is selected based on quantitative performance and inference efficiency. The resulting detections are then processed by a rule-based expert system that assigns maintenance priority levels. This modular separation supports transparency, reproducibility, and independent evaluation of both components.

In short, we want an approach that is simple to repeat, simple to comprehend, and, in fact, is easy to apply. We describe precisely how we divide the data, what are the protocols we adopt, and we maintain it throughout all the runs. Explanations and decision tracing of the deep-learning output are enabled by the rule-based layer at the top of the deep-learning, and that is of paramount importance to infrastructure management. The entire arrangement is a reflection of the actual road-inspection situations, such as poor lighting or poor weather. The assessment is the accuracy of the detector as well as the ability to survive with the various types of damages. We make it as easy as it can be without over-engineering, and the technology can actually be applied in the field. Overall, the approach is evaluated both scientifically and with practical deployment considerations in mind.

5.2 Datasets

This study uses the Road Damage Detection 2022 (RDD2022) dataset, an open and large dataset, which was formed to detect road surface damage using computer vision methods automatically [94, 86]. The data set is comprised of pictures collected in different countries which represent diverse geographic, climatic and infrastructural backgrounds. The images were taken using amateur cameras in the cars and therefore, they have inherent differences in the quality of the image, lighting, and perspective. This natural variation is important to test the strength of object-detection models that will ultimately be implemented in the real world. Some of the damage classes, which are captured in the data, include longitudinal cracks, transverse cracks, alligator cracks and potholes. Each instance of damage is marked with bounding boxes, which are appropriate in object-detection problems. The breadth and size of RDD2022 makes it a suitable benchmark against which experimental findings can be compared and its public nature fosters reproducibility and transparency.

RDD2022 provides predefined training, validation, and test splits released by the dataset creators. We did not manipulate these splits in the course of the study in order to have our

methodology equally related to the past research works [15]. The tuning of the models was done using the training set, the validation set was used to compare and choose the models and the test set was reserved until the final assessment. Such strict division prevents information leakage between stages. The preservation of original splits minimizes bias which may arise due to custom sampling and it also allows us to make direct comparisons of our findings with those of other papers using the same benchmark. This design is a best practice in computer-vision benchmark research.

The annotations of RDD2022 were official and not manually corrected and relabelled. Such a decision retains the original quirks and flaws of the dataset, which may be seen as a representation of real-life situations. The use of the official labels also prevents us to bring about subjective biases that would otherwise occur in case we change the annotations. It makes sure that discrepancies in performance result as a consequence of the models and not modifications to the labels. Data structure and names used in labeling were used consistently throughout all experiments. Any corrupted pictures or lack of tags were automatically filtered rather than being curated manually. In total, our empirical foundation of testing road-hazard detection systems under realistic conditions that are reproducible would be achieved with RDD2022.

5.3 Data Preprocessing Steps

To make everything reproducible and consistent before starting to train and test the models, we configured a data preprocessing pipeline. Because the RDD2022 dataset already provided us with the appropriate annotations of object detection, we did not need to recalculate the labels and manipulate the data at all [15]. Such a decision makes the benchmark hold true and does not introduce any personal bias on the data. All preprocessing was performed the same way on all three sets training, validation, and test to ensure that nothing is bleeding between the various experimental phases. The pipeline did not perform heavy data gymnastics; it primarily performed validation, normalization and input standardization. This conservative strategy is based on the common best practice in deep learning research using benchmarks [53]. It also implies that the variation in performance that we observe is largely because of the architecture of the model and the training configuration. In general, the preprocessing phase maintains a transparent and reproducible condition of things.

Each input image was resized to a fixed resolution of 640×640 pixels, which corresponds to the default input size used in YOLO-based training pipelines [21]. The aspect ratio was preserved via padding to avoid geometric distortion and keep spatial patterns consistent across samples. The training framework automatically scales the bounding box coordinates to the resized images. This resizing technique is a compromise between computation cost and the amount of detail sufficient to identify small and large defects. It is also simpler to compare the variants of models using a standard input resolution, and everything can be fit even to pre-configured network architectures. The same input size allows training to be consistent and give reproducible evaluation outcomes.

The training, validation, and test splits which accompany the RDD2022 dataset remained fully unchanged during the study. This makes our findings as comparable to other literature, and prevents the bias that may arise due to custom resampling [94, 53]. The validation subset was only used to compare models and optimize hyperparameters. The test subset was also not touched during the development process but was rather reserved to the final performance check. This split strategy follows standard machine-learning experimental design and prevents information leakage [53]. It ensures that the scores that are reported actually indicate the quality of the model in generalizing. The retention of

the original dataset splits further enhances external validity of our results.

To identify corrupted images, broken bounding boxes or image-annotation pairs, we performed automated dataset integrity checks. Any samples that did not pass these tests were discarded during training and evaluation to avoid unstable optimization and unreliable metrics [53]. There were no manual corrections to make in this process to make things objective. These cleaning checks eliminate the noise due to bad data without disturbing the data distribution. Constant annotations are particularly important when it is necessary to work with small or blurred objects during object detection. The checks that we deployed allow training to remain stable and the metrics to remain credible. This point of quality control will strengthen our entire pipeline of experiments.

In training we adhered to the classic methods of data augmentation that are already provided by the detection framework, and we used them every time in all the runs. The augmentations were random horizontal flipping, scaling, and color jittering all the mainstays that enhance robustness to various environmental variations [95, 96]. These augmentations resemble variations in lighting, camera positions, and surface appearance that you would encounter on the street in real life when inspecting the road. The augmentation parameters were fixed in all experiments so that we could make a fair comparison of variants of models. We did not incorporate job-specific or handcrafted extensions. This careful policy of augmentation has ensured that the overfitting to artificially created patterns remains low. Latest research indicates that moderate augmentation improves generalization without adversely impacting the evaluation fairness [97].

5.4 Experimental Settings and Software Used

In our study, we established a reproducible and carefully controlled computational environment that allows comparing different model settings and model training runs fairly. Our deep learning pipeline is based on PyTorch that provides us with fast tensor operations and the flexibility required by modern object-detection architectures [98]. The Ultralytics YOLO framework was used due to its standardized implementation, excellent documentation, and high reproducibility assurances, rather than other types of training and evaluation, which are not implemented as widely and do not have comparable conditions of success [99, 21]. This all ran on a Linux machine so as to maintain dependency resolution and runtime behaviour. Library versions were fixed across experiments to prevent silent changes and ensure reproducibility. We stored all the training artifacts, like config files, logs, model checkpoints, to be able to reverse-engineer all of them and debug in future in case something strange occurred. The first piece of information is that under this arrangement, any variation in performance that we observe may rather be attributed to the model or its settings, and not to random environmental noise.

The main training was done on GPU enabled systems whenever we were able to do so, but the preprocessing and expert-system verifications were also made CPU friendly. Such a combination is indicative of the reality of the world: expensive machines are not necessarily on hand to inspect big portions. We have fixed random seeds to load data, initialize models, and apply data augmentation, which is recommended to support reproducibility [53]. The inputs resolution, the way batches were constructed and the way the dataset was divided were also made consistent across all the variants of the model so that the only parameter that could vary was the model or hyperparameter of the model. The training was done using normal gradient based optimizers to maintain consistently stable convergence [100]. We monitored loss curves and validation metrics real time in order to spot any indicators of instability or overfitting as they occurred. Lastly, we also tested all the models on the same hardware and runtime conditions in order to make the comparison meaningful.

The held-out test split of the RDD2022 dataset, to which we did not ever provide any information throughout training or hyperparameter selection, was used as a measure of performance to ensure there is no leakage or optimistic bias. [101]. Common object-detection metrics: precision, recall, and mAP were used by us, which allows us to compare ourselves with other studies. Evaluation was done using the same preprocessing and confidence thresholds in order to make things equal. All the logging was done automatically to give the results a transparent look and debug any problems that arose [102]. Invaluable insights such as final metrics, per-class behaviour and representative detection examples were left to be inspected later through qualitative examination. In the case of comparing architectures we held the hyperparameters constant, surprising the influence of network size and design decisions. This rigorous work process makes sure that whatever performance claims we make can be supported by repeatable and consistent measures. All in all, the experimental set up provides a good base behind the analysis that we provide in Chapter 6.

5.5 Network Architecture

To pick the object detectors we wanted to use in our class project we settled on the YOLO family of one-stage object detectors since they are extremely popular with the real-time work. The key advantage of YOLO is that it performs both localization and classification, thus, the latency is reduced, and it can be deployed to the field more easily than other approaches [21, 103]. YOLOv8 is a good choice since we have to process tons of photos in a short amount of time to scan the road. YOLOv8 was selected due to its strong performance and standardized implementation in the Ultralytics framework [21]. The structure is quite versatile: it has a backbone, neck and detection head. It is then easy to compare the various versions of the model at equal experimental conditions, which is convenient when we change depths and widths. We are in essence going by the current trends in real-time detection since it is the most feasible in our application.

The most important part is the one that extracts hierarchical features per image. It identifies both low-level textures and high-level semantics that are important in identifying road damage. These are then transferred to the neck where there we carry out multi-scale fusion. The information in different resolutions in the neck is synthesized, and everything is sewed together by it [21, 103]. This is essential to road damage detection, in which case you may see tiny cracks on one hand and huge craters on the other. Multi-scale fusion allows the net to remain conscious of small defects as well as obtain the view of the big picture. The purpose of the neck is to ensure good detection regardless of size and that is why we experience real life gains when we apply this design.

The bounding boxes, confidence scores, and class probabilities are directly the output of the detection head on the fused features. Our anchor-free detection tricks are those that YOLOv8 adheres to [28]. We experimented with different depth/width parameters to observe the trade off between precision and speed. Embeddable models are fast on inference and excellent on edge hardware. On the tradeoff of additional compute, bigger models are somewhat more precise. Ultimately, we have settled on the architecture that most effectively did the job in our test, which was not necessarily the largest one. That factual decision ensures that the network is performance-oriented and practically deployed. The overall outcome is that, there is a good basis to depend on in detecting road dangers.

5.6 Other details for reproducing your methodology or results

In order to ensure that you would be able to completely replicate our approach, we executed all experiments on deterministic and well-documented environments that adhere to the best

practices in the field of ML research as of today [53]. Our data loading, weight init, and stochastic augmentation were fixed on random seeds, and nothing other than the model and training configuration can vary, not random noise. I used the same seed in all variants in order to observe differences, which are influenced by architecture and configuration alterations and not by chance.

Our training/validation/test splits were as in the RDD2022 dataset, and data across splits was not mixed. That narrow gap will ensure that leakage is not allowed to enter and ensure a fair assessment. To the point, we are being extremely cautious in order that the results stand the test of time.

The model definitions, hyperparameters and data paths are all stored in external config files keeping things consistent across runs and reducing manual errors. We store checkpoints after every few epochs to be able to draw a model back at any point and not re-train it, examine its state, or conduct other experiments under the same conditions. The checkpoints and configs are tracked by version and it is easy to follow our position at any point in time. These measures are consistent with the reproducibility principles in and allow any other person to expand or recreate the work.

We used a Linux environment to ensure the runtime behaviour was predictable and dependability in dependency handling. The core libraries, such as PyTorch and CUDA, were fixed to a particular version as to prevent bizarre updates or hardware bugs The training speed and inference were fairly compared within the same run because of the similar use of GPUs. [98]. Our validation set was only used to compare the models and to tweak the hyperparameters but the test set was reserved until the last run. This is in accordance with the conventional experimental design in deep learning studies [53]. Even our metrics have been trained under the same conditions hence you will trust the numbers and replicate our results.

5.7 Techniques for Data Analysis / Visualization

In this lab project, we adhered to the general best practices of the deep learning literature to make the analysis organized and the reporting clear [53]. This was aimed at providing us with a good understanding of how each model responded to training as well as its ability to identify the various types of damages in each of the experimental settings. We combined quantitative data with qualitative graphical analysis to make the results easier to interpret and to identify any anomalies that we would not be able to identify otherwise. In order to ensure that the comparison was fair, we followed the same steps of analysis of all the model variants and we created the visual results directly of the logged data to maintain traceability. This has enabled us to maintain the entire process as reproducible and prevent over-subjective decisions. Our reflections lead to the chapter of evaluation which is the next one.

Training dynamics looked at the loss curves and traced the movement of our measures across each epoch. That will allow us to observe whether the model is converging well, training is not converging, or we begin to overfit. Visualization of training and validation loss is a popular technique in teaching how to identify optimization problems of neural nets early enough before they develop into serious problems like overfitting or underfitting the data being fitted to signs of neural net problems. [53]. We automatically tracked all the metrics of our experiments through our experiment-tracking systems to ensure that everything was unchanged across the runs [102]. These diagnostics allow us to identify the bad training behavior at an early stage and adjust the hyperparameters. The observation of the learning curves also enabled us to know the duration of training and when to quit prematurely. Such analysis is especially important during the comparison of lightweight

models with their larger counterpart. The trends that we had identified influenced our decision regarding the models to retain and develop.

To evaluate detection performance, we looked at standard object-detection measurements in form of precision-recall curves and also through class-wise breakdowns. Precision-recall is more detailed than a single metric, particularly when our detection problem is skewed in one direction or another direction [104]. We also examined confusion matrices on the level of classes to identify where the model was confusing one category of road damage with another one. Such an analysis helps to analyze the limits of this model qualitatively and the blurred boundary between classes in a qualitative way [101]. To evaluate the performance of localization and false-positive, we manually inspected the predicted bounding boxes to assess the quality of the prediction. The visualizations were all created using established scientific plotting libraries to help things to be easy to understand and reproduce [105]. Visualization scripts were placed under version control with our code, meaning that we could recreate the figures directly out of the raw results as we needed.

5.8 Evaluation Metrics

In this project, I evaluated the effectiveness of my road hazard detection system by evaluating the typical object-detection measures that the research typically uses in papers in the computer-vision field [48]. These measurements enable me to objectively test both the accuracy of the system to place the hazards and the ability to classify that hazard to a degree that I can make comparisons between alternative versions. Every assessment was carried out exclusively on a held-out test set, thus nothing about training or validation is carried in and keeps the experiment honest as per the advice given in machine-learning research studies [53]. I ensured that I used the same evaluation conditions in all the model variants to ensure that the figures are the same thing every time. The selected measurements are based on the academic standards and practical demands of road-maintenance teams. When combined, they provide me with a clear and comprehensible picture of the quality of good the detections are.

My main quality indicators were precision and recall. Precision is a measure of how relevant the model is at not making a false alarm, the proportion of predicted hazards that are actually hazards. Recall is the percentage of actual hazards that the model is able to identify and it provides an intuition of how many damages are we actually capturing [104]. Finding a balance between recall and precision is particularly difficult in a road-hazard environment, as high rates of false positives will result in unnecessary maintenance repair, whereas false negative may be life-threatening. To analyze this trade-off, precision-recall curves were computed across confidence thresholds. This is a better curve to visualize compared to a single number particularly in case the data is unbalanced.

Intersection over Union (IoU) was used to estimate the matching of the bounding boxes to the ground truth [106]. I viewed a detection as correct when its IoU exceeded a predefined threshold. In this research, the IoU threshold was set to 0.5, which is commonly used in object-detection benchmarks [106]. Then I used the Average Precision (AP) of each of the classes to determine performance at different confidence levels, and averaged the resultant APs to obtain the mean Average Precision (mAP) [106]. I also tried with stricter thresholds of IoU in order to determine the strength of the localization. This provides a more detailed picture of the quality of detection which is used directly to prioritize maintenance tasks [107].

5.9 Hyperparameter Configuration

In our experiments, we tried to keep everything as controlled and reproducible as we could, and therefore we put all hyper parameters in a defined form prior to making any model comparisons. The training parameters remained constant in all settings unless we were trying to test a new experimental condition. In this manner we would be able to isolate architectural variation in a way that there are no confounding influences of hyper-parameter tuning. We used the default parameters provided by the authors of the YOLOv8 modeling framework when training it - schedules of learning-rate, batch building, optimizer dynamics, and so on. [21]. We did not do a large set of hyper-parameter sweep as we believed that would bias the results and complicate the process of reproducibility. We instead maintained a constant baseline and used them in all the runs. Such a modest style would be appropriate to the aim of a level-headed, controlled comparative study as opposed to pursuing crude benchmark results.

Table 5.1: Final hyperparameter configuration used across experiments

Parameter	Value
Input resolution	640×640
Batch size	16
Optimizer	AdamW
Initial learning rate	0.01
Weight decay	0.0005
Number of epochs	50 / 100 (varied per experiment)
IoU threshold	0.5
Confidence threshold	0.25 (default baseline)
Data augmentation	Standard YOLO augmentations
Random seed	Fixed across runs

An input resolution of 640×640 was selected as well to strike a balance between speed and quality when it comes to identifying small defects. That would make the inference faster but would be more likely to conceal fine crack patterns, whereas making it higher would consume memory and time without necessarily improving accuracy. We chose the batch size to approach full utilisation of the GPUs and at the same time maintain the gradients steady. Our optimizer of choice was AdamW since it has a reputation of converting well in detection tasks. We had standard values of learning rate and weight decay that you will find in many object-detection papers [98]. These monolithic parameters were held constant in order to ensure that any variation in performance that we observe is due to the model architecture itself.

5.10 Experimental Workflow Overview

Our experimental workflow was developed as a step-by-step procedure beginning with the data ingestion up to the final assessment. It begins by cleaning and verifying the data to maintain our annotations sound and input normalization clean. After the preprocessing of the images, we train multiple sets of model parameters using exactly those same hyperparameters. We monitor the performance of validation in order to notice when the models are converging well and ensure that they are not over-fitting. The model which best passes the validation metrics is selected and the model is subsequently tested on the hold-out test set. The outcomes of the detection are sent to a rule-based expert system that does

scoring/prioritization of the severity. Lastly, we examine the findings in a quantitative and qualitative manner to determine the extent to which the entire system is performing.

This has made it very easy to separate the development and evaluation phases which makes it to be very organized. The validation data is only used to select the model to ensure no optimistic bias is created. The test set remains unexamined until we have a prepared report on final results. The expert system does not act on the detected results but remains modular in keeping perception and decision making independent of each other. This design is in line with reproducible research principles in applied machine learning [53]. Such a clear definition of each stage is useful to make the methodology transparent. Overall, the workflow provides a good foundation of interpretable and fair experiments.

5.11 Methodological Constraints

Although we paid much attention to the experimental design, there are certainly certain limitations that should be mentioned. We are wholly using the RDD2022 dataset, so we could be missing several other varieties of road surface damage out in the field. The lopsidedness of the types of damage may cause a hiccough in the sensitivity of the model and render the results of the recall somewhat capricious. To maintain the comparable nature of things, we maintained the original train/validation/test splits, but cross-validation was not done due to the inability of our computers. Hardware limits also restricted the size of our batches and the amount of time we could train, and we probably would not have had an opportunity to explore larger model configurations. Also, we did not run a thorough hyperparameter search, so we gave more importance to reproducibility than to the absolute best metrics.

One disadvantage that has been noticed also is that we are using bounding-box annotations rather than pixel-level segmentation masks. Detection metrics are useful, but bounding boxes do not fully capture the irregular geometry of crack shapes. The rule-based parameters of the expert system were as well not learnt based on data but manually tuned. Although it simplifies the system, it may fail to fit into very different deployment environments. The limits of these limitations do not nullify our results, but they place limits on what they can inform us. These limitations should be recognized in order to be a responsible reader of our results and to direct the further course of the study.

6 Experiments and Results

6.1 Model Training

To investigate the influence of four training variations on YOLOv8 performance, we established a controlled experiment in this project. In EXP1 we use the default YOLOv8n. EXP2 introduces additional epochs to understand whether it can obtain a cleaner detection by training longer. EXP3 increases the input resolution to check the ability to detect small defects on large scales. EXP4 replaces YOLOv8n with YOLOv8s to provide the model with additional capacity, but retains other aspects unchanged. This isolates the effect of each change by altering only one variable at a time. Once we examine the standard detection measures and make a visual sanity check, we select a suitable configuration and pass it to the expert system for further analysis. This technique maintains the comparison clean and reduces noise from the data or training environment.

Each run uses the same dataset splits, preprocessing, and evaluation to ensure reproducibility. Our training split is used to train and the validation split is used to measure performance and determine when to stop training. Hyperparameters such as the optimizer, batch size and data augmentation remain fixed unless otherwise mentioned by the experiment. We maintain checkpoints to track training progress. We then evaluate on a held-out test set that was not used during training or model selection to ensure that we do not get too optimistic results. The head-to-head detailed outcomes are presented in Section 6.2.

6.1.1 Environment

The training was established in a highly constrained computing setup to ensure that the optimizations were predictable and consistent. PyTorch was used to construct the entire pipeline providing quick tensor operations and significant support of the current neural network architecture. The majority of the experiments were performed on GPU-enabled machines, to accelerate things, although we used the CPU fallback just in case of the more auxiliary steps. The underlying OS was Linux as it is more stable in maintaining the runtime behavior and dependency management throughout the runs. We have pinned the basic libraries and framework versions so that nothing will go amiss when we update it or when we have an incompatible dependency. We also manually planted random seeds to ensure stochastic processes such as weight init and data shuffling did not have any unexpected behaviour. This setup enables repeatable training runs and supports attributing performance differences to model and configuration decisions.

6.1.2 Training Process

In our supervised learning we worked with annotated road images of the RDD2022 dataset. The aim was to train the models to estimate the bounding box coordinates, objectness, and class of the various types of road damages by the YOLOv8 training pipeline. The training split was only used to optimize the parameters whereas the validation split was used to monitor progress and select the most suitable model.

We resized the input images, as per the preprocessing that we described in Chapter 5. All the important training parameters, such as learning-rate schedule and the batch size, re-

mained the same in all the setups to ensure a fair comparison. We stored model checkpoints after every few epochs because this way we could replicate the results and do subsequent analysis. Lastly, we selected the most suitable model based on validation performance and evaluated it on the held-out test set.

6.2 Experiments

This section describes the experimental design used to test the proposed road-damage detection pipeline. The experiment has been designed to be a set of controlled stepwise settings where one factor was varied at a time. I aimed to find out the influence of training length, input resolution and model capacity on the detection performance where other variables have been fixed. I conducted all experiments on the same splits of data, and under the same preprocessing procedure as well as evaluation procedures to ensure the outcomes would be fair and reproducible. It was focused on separating the effects of each of the factors involved instead of searching through all potential combinations. This assists in giving us a clear understanding of the variation of the performance in the various settings. The quantitative findings of these experiments can be discussed in the subsequent parts of this chapter.

The review covers four setups (EXP1-EXP4) which were designed to isolate the effects of training length, input resolution, and model capacity on a coherent experimental framework. Besides more general measures of detection, I also added some qualitative analyses to verify the behaviour of the model at varying confidence levels. Lastly, I fed the result of the selected detector as an input to the rule-based expert system, which enabled me to test the end-to-end prioritization given a realistic deployment constraint. This allows tracing of any differences in performance to the controlled factors and can be interpreted at the system level which is not limited to the raw measures of detection.

Table 6.1: Overview of controlled experimental variations across EXP1–EXP4.

Experiment	Model	Main modification
EXP1	YOLOv8n	Baseline configuration
EXP2	YOLOv8n	Training epochs: 50 → 100
EXP3	YOLOv8n	Image size: 640 → 800
EXP4	YOLOv8s	Higher-capacity model vs baseline

Table 6.1 summarizes the controlled variations across experimental configurations. The quantitative performance results for these configurations are summarized in Table 6.2.

6.2.1 Comparison of YOLO Configurations

Four experimental setups were conducted to determine which model of the baseline would perform best with the expert system. It concentrated on the influence of various architectural preferences and training environments on the performance of detection. The setup of each of the configs was to test a single variable with all the other variables held constant. I measured performance using standard object detection measures such as precision, recall, and mean Average Precision (mAP). The findings provide a very clear, quantitative foundation of selecting the best model upon which to proceed. The final results of EXP1 to EXP4 are indicated in Table 6.2. The selected configuration is then used as input to the expert system module.

Table 6.2: Comparison of experimental configurations (EXP1–EXP4) including model settings and performance metrics.

Experiment	Model	Epochs	Img size	Precision	Recall	mAP50	mAP50–95
EXP1	YOLOv8n	50	640	0.6546	0.5817	0.6289	0.3514
EXP2	YOLOv8n	100	640	0.6648	0.5936	0.6416	0.3618
EXP3	YOLOv8n	50	800	0.6427	0.6001	0.6411	0.3513
EXP4	YOLOv8s	50	640	**0.6936**	**0.6044**	**0.6633**	**0.3761**

As the results presented in Table 6.2 make evident, adjusting the model architecture and its setup will not only permit training longer, but it will also assist in doing so. Although EXP2 trained 100 epochs and outperformed EXP1 slightly, the additional training time did not substantially increase mAP50–95, which is an indication that the performance level was pegged early on with our dataset. On the other hand, EXP4 demonstrates that configuration adjustments can progressively increase accuracy, recall, and mAP without a longer training run.

When I take a closer look at the numbers, it is somewhat interesting how the EXP4 establishes a good balance between the true positives and false positives. EXP4, in comparison to EXP3 which increased recall but decreased precision, raises both metrics simultaneously, which indicates more stable localization and a more accurate confidence value. This is especially significant in the context of the subsequent step of prioritization, as excessively high rates of false alarms would overweight scores of severity in the expert system. Making a choice of configuration that maintains both precision and recall is thus highly beneficial in enhancing the resiliency of the following decision layer.

From a deployment perspective: EXP4 provides a decent trade off between its speed of execution, and its precision. Big models may provide small additional gains, at the cost of higher inference time and smaller scalability, particularly on small hardware. As our framework combines detection with rule-based prioritization, rather than simply feeding the raw counts on detection into the model, stable and well-calibrated predictions are more important than getting every final point out of metrics. That is why we selected EXP4, not only due to its superior performance over the others in quantitative terms, but also due to its greater stability, interpretability and best suitability to the design of a hybrid expert system.

Figure 6.1: Normalized confusion matrix for EXP4 on the test set.

Then I examined the normalized confusion matrix Figure 6.1 and that provides us a fairly good perspective of the performance of the model on each of the classes with the configuration that we selected. The majority of predictions fall along the diagonal, which is expected to indicate that the model is fairly effective in categorizing the damage. This is consistent with the better results that we obtained in Table 6.2.

It is observed that there is some confusion between classes which are similar in appearance. This is not so surprising due to the ambiguity of some of the defects. This precision is important to downstream prioritization, since when the model predicts a defect wrong, it can immediately confuse the severity prediction.

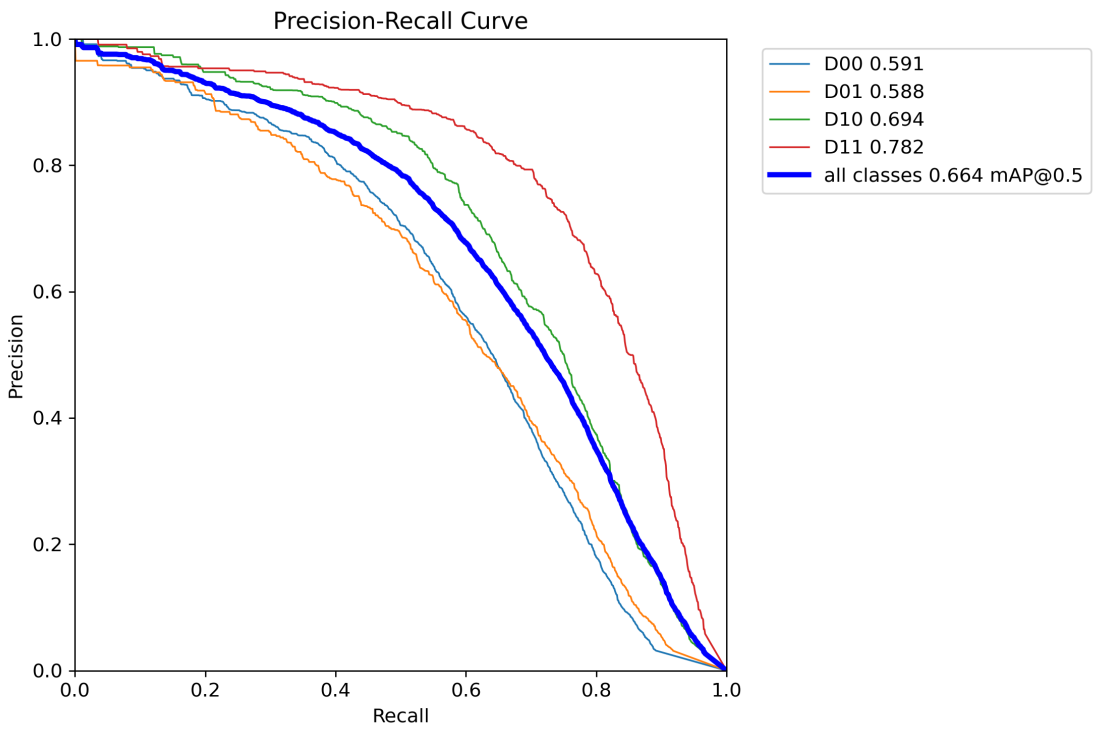

Figure 6.2: Precision–Recall curve for EXP4.

The Precision–Recall curve presents the tradeoff between the recall and the precision at various confidence levels. Figure 6.2 shows that with increasing recall, the model maintains a fairly high precision, i.e. it is capable of identifying more objects without decreasing the precision. This is comparable to the increased accuracy and recall figures that we observed in Table 6.2. The curve is smooth indicating that the confidence calibration is steady and the detections are ranked in a steady manner. My selected model gives a better trade-off between false positives and false negatives than weaker configurations. This is extremely crucial in the real world implementation since there must be good coverage and reliability. It also supports the concept of adhering to set confidence indicators in the application of the model in real-life situations and supports fixed confidence thresholds in practical deployment scenarios.

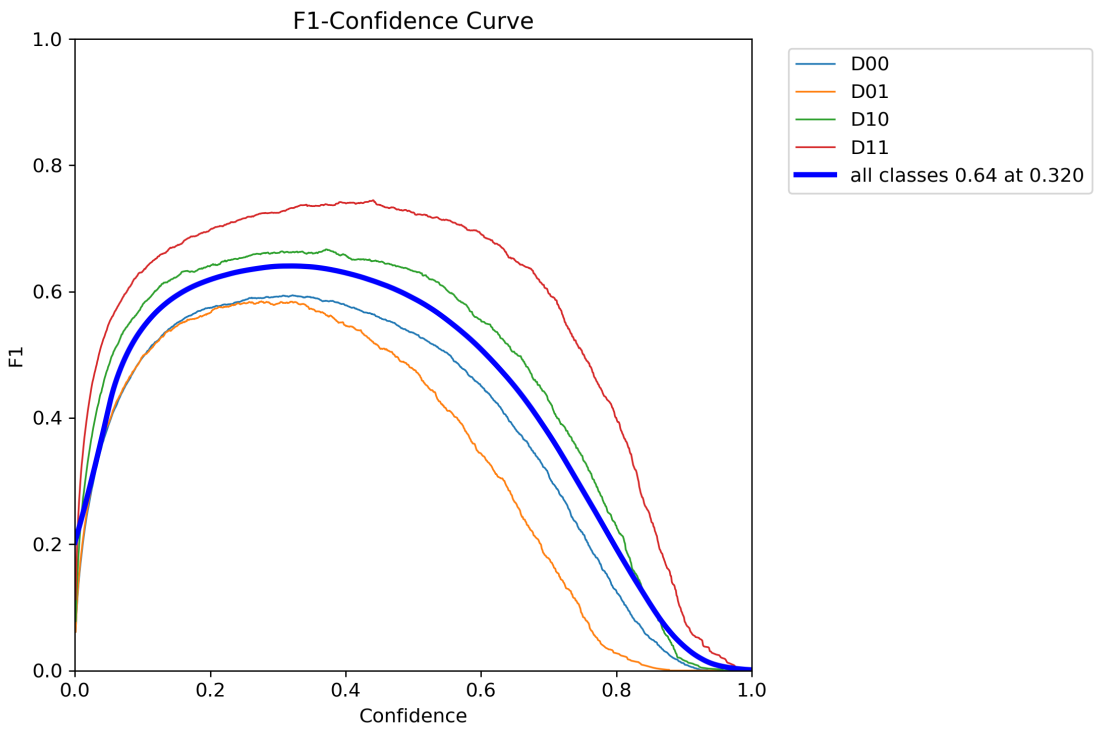

Figure 6.3: F1 score as a function of confidence threshold for EXP4.

The F1 score curve fundamentally looks like the way in which the precision and the recall respond to the variations in the confidence threshold. Figure 6.3 shows that the performance remains relatively stable with a reasonable range of thresholds and this fact shows that the model is not very sensitive to the point we choose to cut-off. The sweet spot is the highest point of the curve which balances the sensitivity on detecting things and the reliability of the detecting things.

This stability enables an easier transition to production, as the threshold will not radically alter things when slightly tweaked. Comparing it to other configs, EXP4 achieves better and more consistent performance and is the evident choice when it comes to the baseline. Essentially, its strength implies that we do not need to be that wary about threshold tuning.

6.2.2 Training Stability and Configuration Consistency

In essence, this arrangement ensures that the randomness is kept in check hence we are able to make comparisons among different set-ups fairly. Random seeds were fixed and the dataset splits were kept unchanged, and all the preprocessing was also done to ensure consistency. We maintained the same training time and batch size and optimizers unless we were intentionally testing these other factors. Under this arrangement, changes in the performance can be explained by anything but random noise. We took care of such issues as the weight initiation and data shuffling. This consistency is extremely critical in interpretation of results when one is making comparisons. The improvement in EXP4 is largely due to the model design, rather than due to oddities of training. This all demonstrates that the experiment was sound and the results that are presented are reliable.

6.2.3 Expert System Integration Experiments

In addition to the detection accuracy measurements of the raw detection accuracy, we conducted additional experiments to observe how the deep-learning detection module performs with the rule based expert system. We are considering the way the outputs of the detection are converted to maintenance priority levels using the preset decision rules. This is not to turn detection accuracy up to eleven; rather we want to verify that prioritisation process remains consistent and interpretable. This demonstrates the usefulness of the combination of data-driven perception and rule-based reasoning. Prioritisation and detection should be kept distinct so that we can appraise each of the parts on a modular basis. These experiments are real-life operational decision-making conditions.

Section 6.6 defines the expert system design, calibration procedure and decision rules.

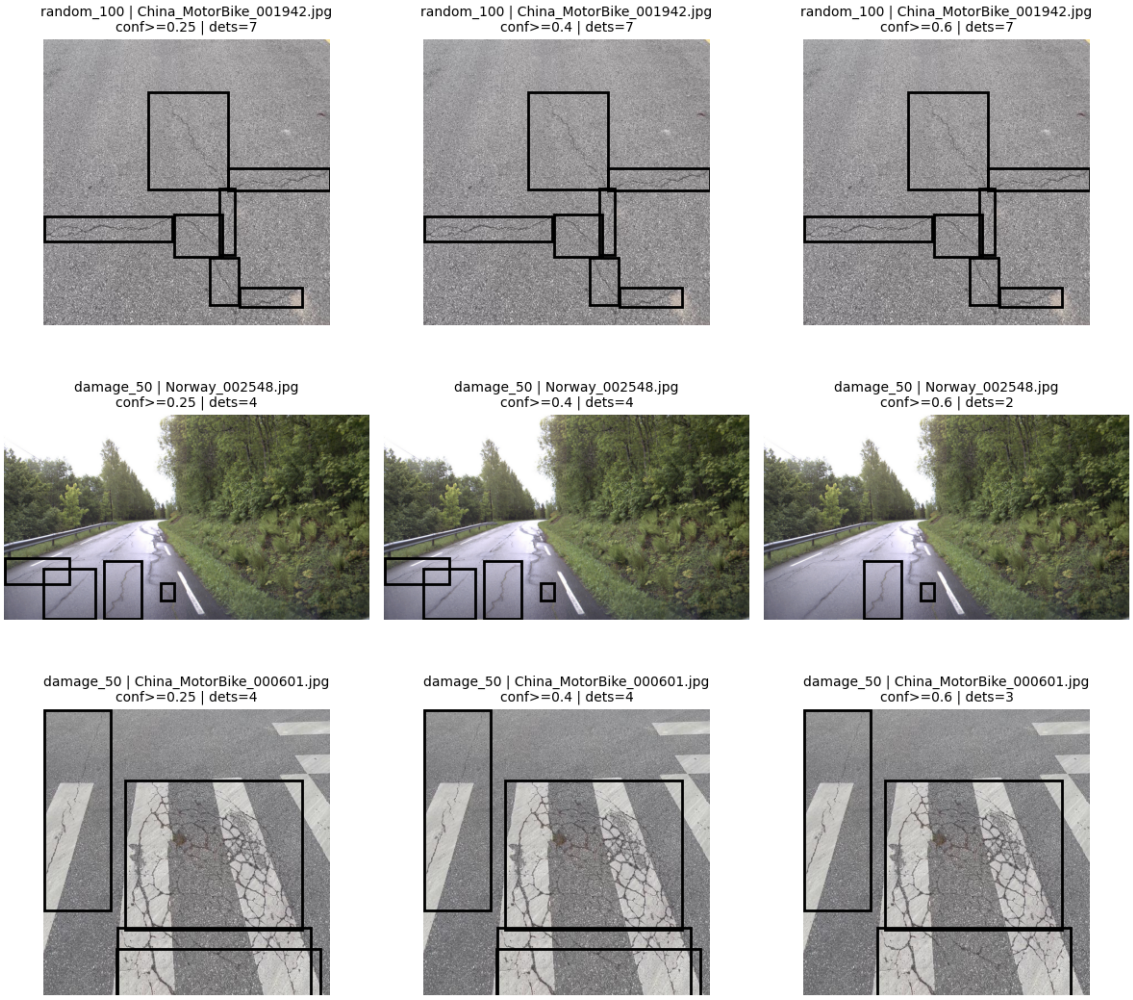

Figure 6.4: Qualitative comparison of detection outputs under different confidence thresholds (0.25, 0.4, and 0.6). Lower thresholds increase recall at the cost of redundant detections, while higher thresholds improve precision but suppress low-confidence predictions.

Figure 6.4 and Table 6.3 highlight the sensitivity of the detector to confidence threshold selection. Lower thresholds increase the proportion of images with detections and the number of predicted instances, improving recall but amplifying redundant and spurious boxes. Higher thresholds reduce the detection density and suppress low-confidence predictions, improving precision but increasing the risk of missing small-scale defects. These obser-

Table 6.3: Sensitivity of detection outputs to confidence threshold selection across evaluation subsets.

Subset	N	Images with det			% images with det			Total det			Avg det per image		
		0.25	0.40	0.60	0.25	0.40	0.60	0.25	0.40	0.60	0.25	0.40	0.60
damage_50	50	46	43	35	92%	86%	70%	113	78	48	2.26	1.56	0.96
random_100	100	66	51	34	66%	51%	34%	142	86	45	1.42	0.86	0.45

vations motivate separating perception from decision-making and delegating operational prioritization to the rule-based expert system rather than relying on aggressive threshold tuning alone.

6.3 Failure Case Analysis and Error Discussion

Although the selected EXP4 setting provided the best total scores on detection, on closer examination of the results, there are a few ways that it still fails, and that is highly applicable in case we ever want to implement it in the field. In places where there are strong linear characteristics such as lane markings, road joints and the edges of shadows we observe a lot of false positive. These patterns resemble a bit cracks to the model and thus the model continues to misclassify them despite the geometry filters. We also observe that small surface artifacts which are not actually a concern can be picked up with moderate accuracy since the convolutional feature extractors respond to fast contrast variations. Many of them are minimized by post-processing, although some noise remains apparent when we lower the confidence threshold. It is important to identify and comprehend these failure modes to determine the level of reliability that the system actually possesses, and it is also the reason why a rule-based reasoning layer is required in order to identify foreseeable perception errors.

The false negatives primarily appear when the cracks are extremely small, when the defects have a low contrast and when the damage is partially concealed. Here the bounding boxes may be invalid or potentially suppressed by non-maximum suppression during the competition of multiple overlapping predictions that vie against one another. Extreme perspective distortions, motion blur, and low light also decrease the level of detection confidence. These issues are the common culprits as far as object detection in scale and challenging imaging conditions is concerned. Besides that, the dataset has class imbalance and therefore the features learning of the rarer classes is weakened by the under representation of certain defect types. This leads to some defects being missed in a systematic manner. Thus, some portion of the detection performance bottleneck is a matter of data, rather than a model per se. That is the reason why the quality of the datasets and the balance of the classes is decisive to the quality of results.

When viewing the confusion matrix by class, there is a low level of cross-class confusion, though there is a visual similarity between defect types. Linear cracks and branching cracks are more frequently mixed as they have almost identical textures. That implies that the acquired representations prefer similarity of texture to higher-level structure. Although the overall accuracy and recall remain fairly high, this confusing can confound downstream prioritization when defects of various types are weighted by different levels of severity. This is addressed in the proposed expert system through the inclusion of geometrical constraints as well as aggregation logic not only on the labels of the classes. Thus, small discrepancies in classifications do not have a significant impact on image-level

prioritization. The perception limitation can be counteracted by an intelligent system design, as illustrated.

The sensitivity of the detector to confidence thresholds brings out the archetypal trade-off of recall and precision. Reduced thresholds increase coverage and detection density but more duplicates and false alarms. A more strict threshold removes low-confidence detections, improving accuracy but increasing false alarms of small cracks. Those dynamics can be observed in Table 6.3 with the significant change of the stats between subsets. Instead of simply adjusting the threshold, the suggested framework combines evidence and then applies quantile-based prioritization to stabilize prioritization. It minimizes instance-level variance and provides more reliable image-level decisions, which is essential when running fixed-scale inspections and can afford to manually optimize thresholds on a case-by-case basis.

On the whole, the examination of the failure case indicates that the vast majority of the occurrences can be attributed to structural causes and are in part manageable through deterministic filtering and aggregation. The rest of the issues are mostly minor defects, classes that are hard to achieve, and imaging conditions. A further increase in data, improved annotations and more focused augmentation would likely advance performance further, although the existing system represents a good trade-off between accuracy, interpretability and the cost of computation. These results further demonstrate the fact that system-level analysis is more insightful than analyzing aggregate detection metrics. Failure analysis incorporation thus enhances the usability of the proposed approach in terms of validity and reliability.

6.4 Model Generalization

Generalization is, in general, the concept that a trained model can still be effective on new data that the model has not encountered previously. In our laboratory, we ensured this on the basis of having a special test set, which we did not touch either during training or validation. In that case we would actually be reporting how well the model is able to deal with new samples and not the extent to which it recalled the training set. Clearly dividing training and validation and testing, we minimize the risk of information leaking or receiving too much optimistic results.

We performed each experiment in the same split and preprocessing, allowing us to make comparisons. This is because the models are not over-fitting because the validation and test performance are in line. Namely, the EXP4 setting is consistently accurate in precision, recall and mean Average Precision, implying that it is learning meaningful visual cues about road damage rather than noise or data shape. The importance of stability to the downstream priority system is related to the inconsistent occurrence of detections, which would negatively affect the expert system.

Nevertheless, despite the decent statistics, the analysis is rather limited as we performed it with the use of one benchmark dataset, namely RDD2022. Although it includes pictures of various countries and different settings, it does not exhaust all of the diversity that we would have in the real world. Other road materials, camera angles, weather conditions or instruments can alter the data distribution in a manner that is not reflected in our assessment. The scores are really only telling us the extent to which the model will perform well on these data, rather than that it will perform well everywhere.

The other aspect of consideration is the detection generalization that is linked to the rule-based expert system. The system is stronger than the model because it aggregates predictions and uses hard filters, although, again, it uses the statistical behaviour of the detection outputs. When the distribution is very different, we may have issues in the

performance of the detection as well as the system prioritisation. It would imply that we must be able to run the entire pipeline in different conditions, and not only analyzing the detection model. There is eventual system-level generalisation which is a blend of perception and decision-making.

To address them, the model needs to be tested on other datasets in the future, and domain-adaptation needs to be attempted. It would be more robust to run the model on the data of other geographies and climates. Such approaches as data augmentation, domain generalisation or fine-tuning on specific domains may assist the model to remain robust in the face of changes. In addition, the domain specific information collection and feedback of actual deployments would enable the system to be more reliable in practice.

On the whole, what we are observing is that the suggested strategy is following stable performance in this controlled environment. To apply it in real life we will require additional validation to be sure that it is in fact robust. A combination of controlled tests and a well-formed system design provides us with a good platform to extend the analysis and strain the constraints of generalisation.

6.5 Comparison with Existing Techniques

In this, we will compare our road-hazard detection and prioritization pipeline with the literature methods. We do not decide to say that we have reached a new state-of-the-art but we want to demonstrate how our system can be integrated into the current research trends and practical implementation requirements. Most of the research on road-damage detection primarily aims to determine the level of accuracy with which it can identify defects, but does not focus on the decision-support component that plays an important role in maintenance planning. We will then place more weight on the specific system design and on the system detection. The findings are presented primarily as qualitative and we will include numbers where they would assist in having a balanced and realistic picture of what we have done.

New methods of road damage detection are largely based on deep-learning object detection architectures, in particular, versions of the YOLO architecture and other one-stage detectors [21, 28, 108]. The approach of these methods aims at maximizing the optimal detection accuracy and inference speed and typically, they provide metrics such as precision, recall, and mAP. The performance is just in the middle of what has been reported concerning the RDD dataset [15, 109]. When other people use much time testing architectures, hyper parameters, or ensembles to achieve a small increase in numbers, we chose a controlled experimental design to ensure that things can be reproducible and interpretable. That is reflective on the pragmatic portion of deploying a system, where transparency and consistency tend to win over a few more points on a scoreboard.

The only distinction between our framework and that of the alternative systems is the expert system that is run with rules contributing to prioritization of hazards. The detection of road damage remains purely a perception issue in the majority of the current methods and the results are never correlated with real-life maintenance actions [88, 89]. In comparison, we divided the pipeline into perception and decision-making stages, which communicate with each other. That division has allowed us to trace detected defects to priority levels in a clear and rule based manner to enable people to trace any particular decision to the specific rule and scores that generated it. This type of design is compatible with the recent push toward explainable, decision-oriented AI, particularly in areas where safety matters, such as infrastructure monitoring [79, 47]. Our system, therefore, provides us with real value of operation more than just detecting numbers by organizing the decision process.

Computationally, the YOLOv8 models we used are as fast as any other real-time detection that uses mobile or low-resource settings [107, 84]. Admittedly, the larger models are providing you with better detection, however, with increased compute requirements. Due to the tightness of the experimental setup (one variable at a time to fit in the model) we can quite clearly see what is driving the differences. That is unlike much of the existing body of research that sets too many knobs simultaneously, and it is difficult to tell where the profitability stems. This is the reason why our systematic analysis provides a more accurate comparison and helps anyone make his decision on how to implement the system in the field.

All in all, our design has its merits in terms of detection performance but introduces a new explicit prioritization logic and a reproducible system design. We did not develop a whole new detection network: we have simply made a nice combination of perception, evaluation and decision-support. That high-level perspective is the one which road maintenance authorities are concerned about, in practice, explainability, robustness, and reproducibility are more important than raw accuracy. Consequently, we consider our framework to be a balanced and practically applicable solution to the existing research on road hazard detection.

6.6 Rule Based Expert System for Hazard Prioritization

In this section, we present the rule-based expert system developed to convert the raw outputs of the detector into maintenance priorities. Although modern object detection models can identify surface defects with high accuracy, they do not directly provide information suitable for real-world maintenance decisions. Raw bounding box predictions often include false positives, fragmented detections, and spatially irrelevant responses. Therefore, an additional reasoning layer is required to bridge the gap between defect observation and maintenance decision-making.

To address this, deterministic filtering rules and a continuous scoring mechanism are introduced to transform instance-level predictions into a severity score at the image level. The design emphasizes interpretability and reproducibility, ensuring that all decisions can be traced to explicit conditions. Geometric constraints, region-of-interest filtering, and appearance-based heuristics are combined to reduce systematic detection errors. The final output is a prioritization structure that categorizes each image into low, medium, and high severity levels, providing a simple and scalable approach for integrating deep learning outputs into maintenance workflows.

Table 6.4: Postprocess v2 filtering and priority tier distribution on the test set.

Metric	Value
Total detections before filtering	8304
Detections after filtering	5711
Filtering reduction	31.2%
LOW priority images	2121
MEDIUM priority images	606
HIGH priority images	303

The postprocess v2 stage removes approximately 31.2% of detections, including geometrically implausible boxes, lane-marking artifacts, and detections outside the region of interest. This reduction improves interpretability while preserving structurally relevant

defects. The resulting tier distribution shows that only a small subset of images (303) is classified as HIGH priority, allowing the system to focus attention on the most critical cases.

6.6.1 Quantitative Impact of Expert Postprocessing

To evaluate the effectiveness of the expert system, we analyze the reduction in raw detections and the resulting severity scores. The postprocess v2 filter reduces the number of detections by 31.2%, where most removed boxes correspond to noise from lane markings or irrelevant patterns. Instead of removing meaningful detections, the filtering process emphasizes structurally significant predictions.

As shown in Figure 6.5, HIGH-priority samples typically contain large defects, consistent detections, or dense clusters of cracks within the road region. In contrast, LOW-priority images mostly include small or isolated detections with limited spatial relevance. This separation demonstrates the ability of the scoring mechanism to distinguish between minor surface imperfections and more severe structural damage.

The quantile-based tiering adapts dynamically to the distribution of scores within each batch. Instead of relying on fixed thresholds, the system derives priority levels from the observed data distribution, improving robustness to variations in detection confidence.

Overall, combining hard filtering with continuous scoring improves interpretability while ensuring that relevant detections are retained. The expert system transforms raw detection outputs into structured and actionable maintenance priorities.

Figure 6.5: Top ranked HIGH priority samples produced by the expert system (postprocess v2).

6.6.2 Inputs and normalization

The expert system receives detection outputs exported as a structured table. Each row represents a predicted instance, including confidence score and bounding box coordinates. From the predicted corners, the bounding box width w, height h, and center coordinates (x_c, y_c) are computed. These values are normalized by the image width W and height H to obtain $w_n = w/W$, $h_n = h/H$, $x_{c,n} = x_c/W$, and $y_{c,n} = y_c/H$.

This normalization ensures that geometric properties remain consistent across images with different resolutions, which is essential for applying threshold-based filtering. A minimum confidence threshold of 0.25 is applied to remove low-confidence detections.

All thresholds and weighting parameters used in the expert system are calibrated in Chapter 5 and kept fixed during evaluation.

6.6.3 Geometric and Spatial Hard Constraints

In the first stage of the expert system, explicit geometric and spatial constraints are applied to remove implausible detections. These hard constraints aim to reduce systematic false positives while preserving relevant road defects.

Let the detected bounding box have width w and height h. The ratio of the sides is given by:

$$a = \max\left(\frac{w}{h}, \frac{h}{w}\right). \tag{6.1}$$

If a exceeds a predefined threshold $a_{\max}$, the detection is discarded. Formally:

$$a > a_{\max}. \tag{6.2}$$

This rule captures elongated boxes that tend to originate in lane markings or road borders and not necessarily some real structural damage.

To prevent off-road detections, a trapezoidal region of interest (ROI) is defined in normalized image coordinates. Assume that the center of a detection is given as (x_c, y_c) (normalized). We keep a detection only if:

$$(x_c, y_c) \in \mathcal{R}_{road}, \tag{6.3}$$

The trapezoidal region provides a coarse approximation of the drivable road area.

Detections that are excessively thin in either dimension are also removed, they are characteristic features of lane markings or other details. The rejection rules are:

$$h_n < \tau_h \ \wedge \ w_n > \tau_w, \tag{6.4}$$

or

$$w_n < \tau_w \ \wedge \ h_n > \tau_v, \tag{6.5}$$

w_n and h_n are width and height divided by the dimensions of the image.

6.6.4 Color and Shape-Based Suppression

In addition to geometric filtering, simple appearance-based heuristics are used to suppress common false positives.

Lane markings typically appear as elongated white or yellow patterns. We define r_{lane} as:

$$r_{lane} = \max(r_{white}, r_{yellow}). \tag{6.6}$$

Detections with high lane-color response and elongated shape are removed when:

$$r_{lane} \geq \tau_{lane} \ \wedge \ a \geq a_{lane}. \tag{6.7}$$

To handle circular structures such as manhole covers, a circularity measure c is used. Instead of removing these detections, a penalty is applied when:

$$c \geq \tau_{circ}. \tag{6.8}$$

These heuristics reduce noise while preserving potentially relevant structural anomalies.

6.6.5 Detection-Level Scoring Function

Following hard constraint filtering every surviving detection is given a continuous severity score.

Each detection is characterized by its confidence $conf$ and normalized bounding box area A_n. The level of detection is determined as:

$$S_{det} = conf \cdot A_n^{\alpha} \cdot P_{aspect} \cdot P_{lane} \cdot P_{manhole}, \tag{6.9}$$

where:

- $P_{aspect} = a^{-\beta}$ penalizes elongated detections,
- P_{lane} reduces the score for moderate lane-color responses,
- $P_{manhole}$ applies a fixed reduction for circular metallic structures,
- α and β are tunable weighting exponents.

The formulation is such that large high-confidence geometrically plausible detections will have a greater impact on final image-level severity.

The scoring mechanism is based on the assumption that visual features correlate with structural severity. Higher confidence values indicate stronger model agreement, while larger normalized areas correspond to more extensive damage. The aspect ratio penalty minimizes the impacts of the long detections that in most cases are related to non-defect structures like lane markings. In the same manner, appearance-based penalties reduce common false positives without entirely eliminating possibly pertinent detections. This design prioritizes geometrically plausible and contextually meaningful detections. The system features a lack of binary decision boundaries due to the combination of various factors into one continuous score and enables gradual differentiation of the severity of defects. This is especially in the real world where the severity of damage is necessarily continuous and not discrete. The resultant score is thus a more realistic description of the severity of the condition of the roads.

6.6.6 Image-Level Aggregation and Quantile Tiering

At the image level, we aggregate the scores on the detection:

$$S_{img} = \sum_i S_{det,i} \cdot (1 + \lambda \log(1 + n_{det})), \tag{6.10}$$

In which n_{det} is the number of detections we retained and λ is simply a small amplification factor which slightly increases the influence of multiple detections when we make multiple detections.

Rather than determining strict levels, we allow the data to determine the levels of priority by drawing empirical quantiles of the score distribution:

$$\text{Tier}(S_{img}) = \begin{cases} \text{HIGH}, & S_{img} \geq q_{0.90} \\ \text{MEDIUM}, & q_{0.70} \leq S_{img} < q_{0.90} \\ \text{LOW}, & S_{img} < q_{0.70} \end{cases} \tag{6.11}$$

This quantile-based strategy adapts to whatever distribution we actually observe in deployment avoiding manually defined thresholds.

The integration of the expert system enables structured post-processing that reduces false positives while preserving critical detections. The fact that the tiering is quantile-based means that manual threshold tuning is not required which only adds to the overall robustness of the system regardless of how the deployment environment evolves. Overall, combining a data-driven detection model with rule-based logic improves both usability and reliability in real-world scenarios.

Although the rules are deterministic, the resulting tier distribution depends on the score distribution of the dataset. High-severity cases are concentrated in a small subset of images, which is consistent with the relatively low frequency of severe defects in real-world data. Visual inspection of HIGH-priority samples indicates that the system successfully highlights structurally significant damage while reducing noise.

However, the absence of ground-truth severity annotations limits the ability to quantitatively validate the prioritization. Future work should include evaluation against expert-labeled severity scores to strengthen the assessment of the system.

6.6.7 Impact of Expert System on Decision-Making

Overall, the results show that the expert system provides an effective post-processing layer for organizing raw detection outputs into meaningful maintenance priorities. While the object detection model produces instance-level predictions, these outputs alone are not sufficient for decision-making, as they do not provide information about severity or relevance.

The proposed rule-based system aggregates detections into a global image-level severity score, improving interpretability and reducing the impact of isolated false positives. Consistent patterns of detections become more prominent, allowing structurally significant damage to be identified more reliably.

Filtering removes common artifacts such as lane markings and irrelevant detections outside the region of interest, improving the reliability of the prioritization process without discarding critical information.

The quantile-based tiering further enhances robustness by adapting automatically to changes in detection density and environmental conditions. Overall, the expert system improves both the interpretability and practical usability of the detection pipeline, transforming raw predictions into actionable maintenance insights.

6.7 Discussion

The previous sections presented the experimental evaluation of the detection models and the integration of the expert system. However, quantitative metrics alone are not sufficient, the results must be interpreted in terms of practical implications, system limitations, and

real-world deployment constraints. In this section, we summarize the key findings, discuss shortcomings, and outline potential directions for improvement from a system-level perspective. Particular emphasis is placed on the interaction between detection and prioritization, since maintenance decisions depend on both components rather than on raw detection statistics alone. We also discuss limitations related to the dataset and to design choices in the proposed pipeline. Overall, the goal is to provide a realistic assessment of how the system would behave in a deployment scenario.

The experiments indicate that lightweight YOLO-based models can provide a favorable balance between detection performance and computational cost for road damage detection. Rather than optimizing solely for peak benchmark accuracy, stable and well-calibrated predictions are more valuable when outputs are directly consumed by a decision-support layer. The results further suggest that architectural choices have a stronger impact on performance than marginal changes such as extending training by a small number of additional epochs under the same setup. Increasing model capacity yields more consistent gains than minor extensions in training time, which supports selecting an appropriate model configuration instead of relying on extensive hyperparameter tuning. This aligns with practical deployment requirements, where inference efficiency and prediction consistency are as important as small improvements in benchmark scores.

Despite these strengths, several limitations affect both detection quality and overall reliability. Very small defects and minority classes remain challenging, largely due to class imbalance and the limited number of representative training examples. Environmental factors such as illumination changes, shadows, compression artifacts, and camera perspective can also reduce confidence stability and increase variability in predictions. These challenges are common in real-world road inspection settings and reflect known limitations of vision-based one-stage detectors. In addition, annotation noise can influence measured performance, particularly when defect boundaries are ambiguous or when categories exhibit visual overlap. Therefore, part of the observed variation is attributable not only to model behavior but also to dataset characteristics, which should be considered when interpreting per-class performance and failure modes.

The inclusion of the rule-based expert system improves the practical usefulness of the detection outputs by converting raw predictions into structured severity levels. The post-processing stage reduces systematic false positives by applying deterministic filters, aggregating detections, and emphasizing consistent spatial patterns. This is particularly beneficial when confidence thresholds change, since detection density can vary substantially. The quantile-based tiering further stabilizes prioritization by using the empirical score distribution rather than fixed global cut-offs. Importantly, the expert system does not replace the detector; it complements it by mitigating systematic errors through explicit rules and continuous scoring. This interaction between perception and reasoning represents a key strength of the proposed framework.

A further limitation is the reliance on a single benchmark dataset. While results are consistent under controlled conditions, additional evaluation across diverse datasets would be required to assess robustness across geographic regions, pavement types, camera configurations, and environmental conditions. Cross-dataset experiments would also help quantify the sensitivity of expert-system parameters to shifts in data distribution and strengthen conclusions about generalization. Future extensions may include targeted augmentation strategies for difficult defect classes, explicit severity estimation approaches, and deployment optimizations such as pruning or quantization. Incorporating temporal information from repeated inspections could also improve robustness by enabling tracking of defect progression over time and reducing uncertainty.

Overall, the proposed pipeline provides a transparent and modular approach for integrat-

ing object detection with decision support in road maintenance. By combining data-driven perception with deterministic prioritization, the system moves beyond defect identification toward actionable maintenance guidance. The results indicate that effective prioritization can be achieved even when detection performance is imperfect, provided that errors are handled systematically through post-processing and scoring. This highlights the importance of system-level design over optimizing exclusively for peak detection metrics. Although broader validation in diverse real-world conditions remains necessary, the proposed framework provides a solid foundation for practical deployment and demonstrates how hybrid approaches can bridge the gap between vision-based detection and operational decision-making in infrastructure management.

7 Conclusion

This thesis proposes a hybrid framework for automated road hazard detection and prioritization that combines a deep learning-based object detector with a rule-based expert system. The primary challenge addressed is the gap between visual detection and actionable decision-making in infrastructure maintenance. While many existing approaches focus on detection accuracy, the proposed framework extends this perspective by incorporating a structured and interpretable prioritization layer. By separating perception and reasoning, the system achieves improved transparency, reproducibility, and practical applicability.

The experimental results demonstrate that lightweight YOLO-based models provide an effective balance between detection performance and computational efficiency. A controlled study was conducted in which multiple configurations were systematically evaluated to analyze the impact of training duration, input resolution, and model capacity. The findings indicate that increasing model capacity yields more stable performance improvements than extending training duration. The selected EXP4 configuration achieved the best overall results in terms of precision, recall, and mean Average Precision. Qualitative analysis further confirms stable detection behavior and reliable confidence calibration, which are essential for downstream decision-making.

Beyond detection, this work introduces a rule-based expert system that transforms instance-level predictions into image-level severity estimates and priority levels. The system integrates geometric constraints, region-of-interest filtering, and appearance-based heuristics within a structured scoring framework to reduce false positives and improve interpretability. The use of deterministic rules ensures that each decision remains transparent and traceable. The post-processing stage effectively filters noise while preserving relevant detections, resulting in consistent hazard prioritization. The results demonstrate that reliable decision support can be achieved even when detection outputs are imperfect, provided that errors are handled systematically.

The analysis of failure cases highlights limitations related to small-scale defects, rare classes, and challenging visual conditions such as shadows and perspective distortion. These issues are consistent with known challenges in object detection and are partially influenced by dataset characteristics, including class imbalance and annotation variability. While the expert system mitigates some of these issues by aggregating evidence and reducing systematic errors, it does not fully eliminate them. This reinforces the importance of combining data-driven and rule-based approaches in safety-critical applications.

The generalization analysis shows consistent performance across validation and test sets. However, as the evaluation is based on a single benchmark dataset, further validation across diverse environments is required to assess real-world robustness. Variations in road materials, imaging conditions, and geographic regions may affect both detection and prioritization performance. Therefore, the results should be interpreted as in-distribution generalization rather than as a guarantee of deployment-level robustness.

Future work may extend this framework in several directions. Cross-dataset evaluation and domain adaptation techniques could improve robustness under varying conditions. Additional data collection and targeted augmentation strategies may enhance performance on rare and small-scale defects. The incorporation of temporal data from repeated inspections could improve reliability by enabling defect tracking over time. System-level optimizations, including model compression and quantization, may support real-time deployment

on resource-constrained devices. Furthermore, integrating expert-labeled severity annotations would enable a more rigorous evaluation of prioritization quality.

In summary, this thesis demonstrates that automated road hazard analysis can be effectively addressed through the integration of deep learning-based perception and rule-based reasoning. The proposed framework emphasizes not only detection accuracy but also interpretability, reproducibility, and practical applicability. The results indicate that hybrid approaches effectively bridge the gap between computer vision outputs and operational requirements in infrastructure maintenance. The proposed pipeline provides a structured and scalable foundation for future research and real-world deployment.

7.1 Summary

This thesis presents the design, implementation, and evaluation of an automated system for road hazard detection and prioritization based on deep learning and rule-based reasoning. The approach follows an end-to-end pipeline that includes data preprocessing, model training and evaluation, and a decision-support expert system. The main goal is to develop a reproducible and practically usable solution that can detect road surface damage and transform these results into structured maintenance priorities. The work focuses on transparency, modularity, and controlled experiments, instead of relying on very complex model architectures.

To ensure a consistent evaluation, a benchmark-based strategy was used with the RDD2022 dataset. Multiple YOLO-based models were trained and evaluated under the same experimental conditions. The final system combines visual perception with rule-based prioritization in order to connect detection results with practical decision-making. The results show that modern object detection models can be effectively combined with simple and interpretable logic for real-world infrastructure inspection.

The main contributions of this thesis can be summarized as follows. First, a hybrid architecture was developed that integrates deep learning-based detection with a rule-based expert system, helping to connect perception results with maintenance decisions. Second, a controlled experimental setup was designed to analyze the impact of training duration, input resolution, and model capacity on detection performance. Third, a deterministic post-processing and scoring mechanism was introduced to convert raw detections into interpretable severity levels. Fourth, a quantile-based prioritization approach was used to allow adaptive hazard classification without manual threshold tuning. Finally, the overall system was developed with a strong focus on reproducibility, modularity, and practical applicability.

The methodology follows a structured and reproducible process. The work started with an analysis of existing road damage detection approaches and available benchmark datasets. The RDD2022 dataset was selected and prepared using fixed preprocessing steps and consistent training, validation, and test splits. Several YOLO-based models were trained under controlled conditions to allow a fair comparison. Experimental configurations were varied to analyze the effects of training duration, input resolution, and model capacity. All experiments were conducted using fixed random seeds to support reproducibility. The selected reference configuration was then used to generate detection outputs, which were used as input for the rule-based expert system.

The experimental results indicate that architectural choices have a stronger impact on performance than simply increasing the training duration. A balance between precision and recall is important for reliable downstream prioritization. Increasing model capacity improves detection stability and confidence calibration, without the need for excessive optimization. The expert system improves interpretability by applying deterministic fil-

tering, scoring, and aggregation of detections. This helps reduce false positives and makes the prioritization easier to understand and trace back to explicit rules. As a result, the proposed pipeline provides a reproducible and interpretable framework for automated road hazard analysis that can be used in real-world infrastructure monitoring scenarios.

The steps involved in this work can be summarized as follows:

- Analysis of existing road damage detection approaches and benchmark datasets.
- Selection and preparation of the RDD2022 dataset for controlled experiments.
- Design of a modular architecture separating detection and prioritization.
- Training of multiple YOLO-based model configurations.
- Evaluation using precision, recall, IoU, and mean Average Precision.
- Development of a rule-based expert system for prioritization.
- Integration of detection outputs with decision-support logic.
- Documentation of all steps to ensure reproducibility.

Overall, the results show that combining deep learning-based perception with rule-based reasoning can provide a practical and interpretable solution for automated road hazard analysis. The proposed framework achieves solid detection performance while also enabling structured prioritization of maintenance actions. By focusing on reproducibility, modular design, and decision support, the system provides a strong basis for both research and practical deployment in infrastructure monitoring. These findings highlight the importance of linking detection performance with interpretable decision-making in order to bridge the gap between computer vision models and real-world infrastructure management.

7.2 Dissemination

This system for road-hazard recognition and prioritization could be applied in real-world road inspection workflows. The point is that we would be able to connect it to infrastructure checking devices and intelligent transportation systems, and the city developers and repair teams receive automatic inspections that would reduce manual effort and allow them to check the roads more frequently. Integrating object detection with rule-based prioritization provides outputs that maintenance teams can interpret at a glance. It means that it is not an academic project which is only academically interesting, it may also serve to make practical decisions in the field of action. The system is designed in a modular manner, therefore, can be integrated into larger smart-city or infrastructure systems. The detection module is a self-contained perception module and the prioritization layer is capable of changing to suit local maintenance policies and operational limits. It is not merely an experiment in a laboratory but it can even form the basis of real-life implementation.

From a broader research and industry perspective, there is much that this framework can contribute to research communities, in particular, to intelligent transportation systems, computer vision, and infrastructure analytics researchers. By relying on common datasets and reproducible experiments, that assist everybody in comparing new detection methods on a fair level. The obvious division between perception and decision logic also implies that it can be readily extended to activities such as severity estimation, geospatial analysis or multi-sensor fusion. Besides, the expert-system layer provides a transparent decision-making which connects with the impetus toward explainable AI in safety-critical

environments. I find this methodology a strong foundation of further dissemination of academic nature such as a conference or journal paper in the future. The interchangeable environment and benchmark-based assessment also enables other researchers to build upon and confirm our results. In general, the system may be used as both a working prototype and a study of reference in the future research of future projects.

7.3 Problems Encountered

When designing and testing the proposed system, there were a number of challenges that were faced which affected the experimental design and implementation choice. The major challenges were dealing with inconsistency and flaws in the dataset annotations. RDD2022 is a standard benchmark, but due to the imbalance in the classes, small objects, and labelling quality, the imbalance created some difficulties in training and testing. Some categories of damage were much less common than others, and this interfered with the stability of recall and performance per class. These problems necessitated close management of augmentation plans and evaluation processes over and above manual amendment of datasets. This was a limitation to the use of aggressive optimization methods that would be used to artificially boost benchmark scores. It was consequently more important to ensure methodological integrity as opposed to short-term measures of metrics. Consequently, the reported results are realistic data conditions as opposed to optimized benchmark performance.

The other issue was how to balance detection performance and computational performance in terms of different model configurations. Bigger architectures enhanced robustness on localization at the expense of training with higher computational requirements. Lightweight models on the other hand minimized on resource usage but at times had challenges in addressing small scale or low contrast defects. It was necessary to systematically experiment with a controlled sort of experiment to determine an appropriate trade-off. The requirement to be able to compare experiments fairly required stringent similarity in the dataset division, the preprocessing phase, and the training parameters. This brought forth a higher level of experimentation and demanded close management of configuration. Moreover, stochastic effects required control by fixed random seeds and consistent environments in order to obtain reproducibility. These limitations made the process of the experiment more rigorous yet guaranteed credible and interpretable findings.

Another difficulty was the challenge of incorporating the results of deep learning into rule-based logic of prioritization. Conversion of bounding box predictions to maintenance priorities needed clear and consistent decision rules. The formulation of these rules was done through trade-offs between interpretability and expressiveness adequate to accommodate real-world maintenance concerns. Very complicated sets of rules would decrease transparency, whereas very simple sets of rules may omit any contextual information. The last expert-system design is thus a compromise between readability and utilisation. There were also features that could not be incorporated as a result of the scope e.g. dynamic severity estimation and the temporal damage progression. Irrespective of such limitations, the system has a consistent combination of perception and decision-making. This shows that data-driven detection and interpretable reasoning can be used together in one system.

7.4 Outlook

Looking ahead, this project could be extended in several directions. The first step that occurs to the mind is to add a more advanced severity estimation to the case so that we can even know whether the crack is a big issue or a normal wear-and-tear. We are currently

using rule-based logic to classify the type of defects, but had we the ability to estimate the magnitude of the damage using regression, or even perform pixel-by-pixel segmentation we would provide engineers with a much more precise set of maintenance recommendations. That would make the people on the ground see clearly which of the little bumps should be patched and what are the structural problems that are pressing. And on top of that, in case we begin to track the same spot over time we would be able to notice how a defect develops and possibly even know when it will reach a critical point.

Another direction is improving real-world efficiency. Although our current setup is a good trade off between performance and precision, it may be possible to improve efficiency through model compression, pruning, or quantization by compressing the model, eliminating unwanted weights, or quantizing it to consume less power in the edge devices. It is particularly necessary in regard to fleet checks in which the on-board hardware may not be incredibly powerful. Testing on embedded hardware would strengthen evidence of real-time feasibility and scalability. Integrating GPS metadata could also support route-level prioritization by grouping defects by location.

On the decision-making side, we could go beyond the rule-based prioritization. The system can be made adaptive to past data and context using techniques such as probabilistic reasoning, fuzzy logic or even machine-learned decision trees. The integration of those methods and explainable AI would increase the confidence of the automation. Conducting cross-dataset tests would also be useful in our understanding of the model to withstand varying lighting, weather, or road conditions. The more we validate, the better we are going to be sure of the framework.

Overall, this framework provides us with a clean interpretable method of automating the analysis of road hazards. This combination of reproducible experiments with rule-based decision making is creating a system that is reliable and scalable. The findings reflect that the combination of data-driven perception and rational logic could indeed enhance transparency and make the tool more practical for field use. Such explainability is important to infrastructure monitoring, a decision cannot be a black box. Although we have further work to do in the field to test it more, the given set-up is the basis of rolling it out into practice. Future work will focus on optimizing the perception and reasoning layers, expanding coverage to more conditions, and improving reliability in practical deployments.

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